China Journal
2007
Jan and Patricia Miller

Sunday, 9 September 2007

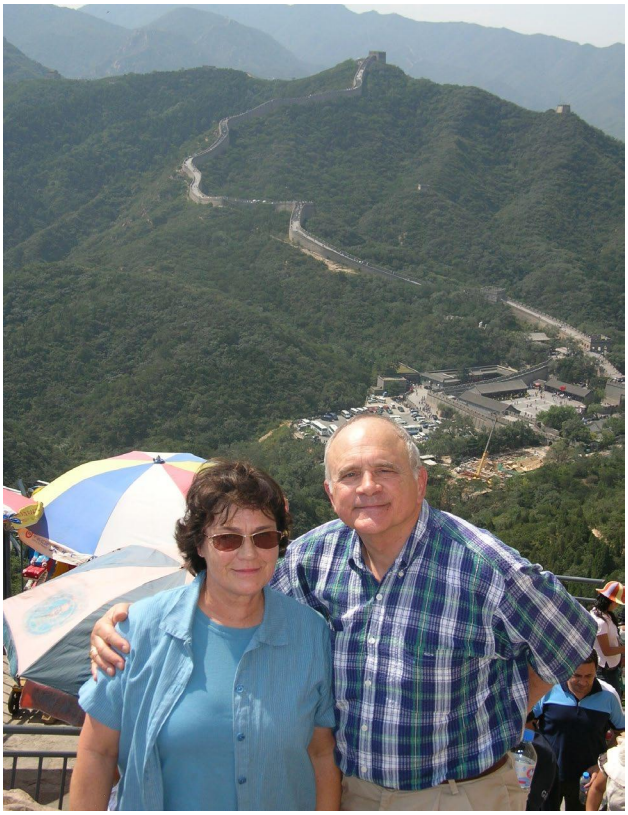
We arrived in Beijing without one bag. Dr. He, Professor Hu Yuehua's assistant, from Central South University, met us at the International Airport and took us to the Xiyuan Hotel. The accommodations were excellent!

Patricia, JDM, and Professor Wang Dianzuo,
Beijing, September 2007

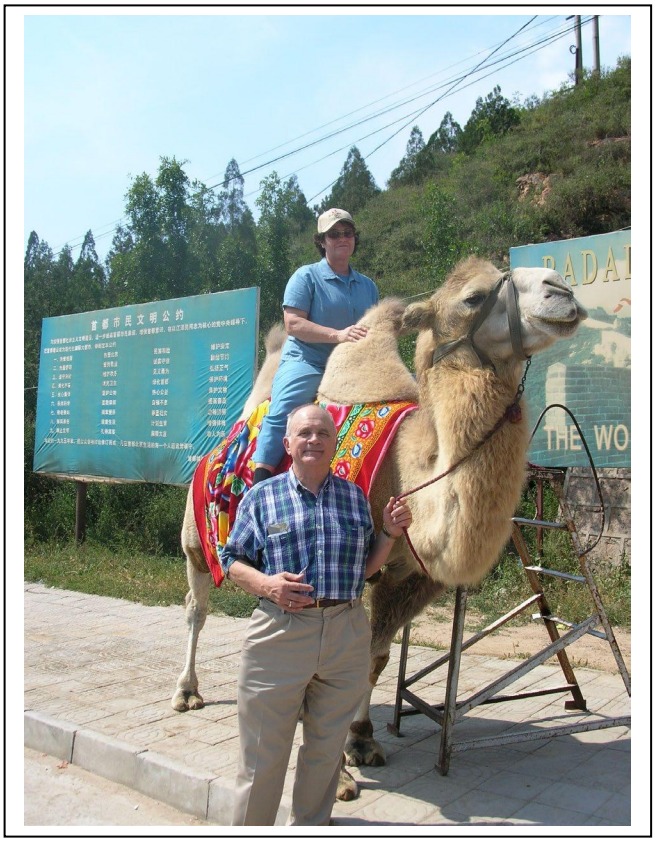
We had dinner with Professor Wang Dianzuo, former President of Central South University and retired Vice President, Chinese Academy of Engineering. After dinner we were treated to entertainment and refreshments at the famous Losche Teahouse where we had Chinese opera, acrobatics, shadow theater, magicians, and many other very interesting performances. In fact Granny even participated from the audience in one of the activities in which the actor was changing faces, quite an amazing performance in which the actor changes his face (mask) so fast that you cannot see the change. The artist came off stage to approach Granny who was in the front row of the audience. Granny was invited to touch the face of the artist and as soon as Granny reached out, the face (mask) changed, essentially instantaneously, so fast that we could not see the change, occur even right before our eyes.

Monday, 10 September 2007

We took a day trip to the Great Wall and the Summer Palace with Dr. Ma from the Beijing Nonferrous Research Institute. At the Great Wall, about an hour or two from Beijing, we had an excursion passing at least four stations on the climb. After which we rode the cable car, gondola, back to the Badaling entry point. Before departing to return to Beijing we had photos with Granny on the camel with Papa being her faithful assistant. After lunch we visited the Summer Palace (beautiful lake and colonnade; visit included boat ride past temple to return to entrance) and then had dinner with Dr. He at the rotating restaurant on the top of our hotel, the Xiyuan Hotel.



Patricia and JDM at the Great Wall,
Badaling, September 2007



Patricia on the camel at the Great Wall with
JDM her faithful assistant, September 2007

Tuesday, 11 September 2007

Dr. Ma took us to the Forbidden City in the morning. If memory is correct the Forbidden City was occupied by emperors of the Qing Dynasty up to the beginning of the 20th Century. We had a marvelous lunch of Beijing Duck at the famous restaurant chain, China Quanjude, established in 1864. We had duck number 443,646 at this location. After a rest in the afternoon, we went to have measurements taken for special silk clothes to be made in Changsha for the lectures and award of Honorary Professor. Prior to the evening meal we were received at the Chinese Academy of Engineering by Professor Wang Dianzuo, Jincheng Kang, Deputy Director General of International Affairs, Dr. He, and others. After dinner we returned to the Xiyuan Hotel. Finally our missing bag appeared and we had fresh clothes to wear. Praise the Lord! We had some concerns as to how we were going to survive with one set of clothes.

Wednesday, 12 September 2007

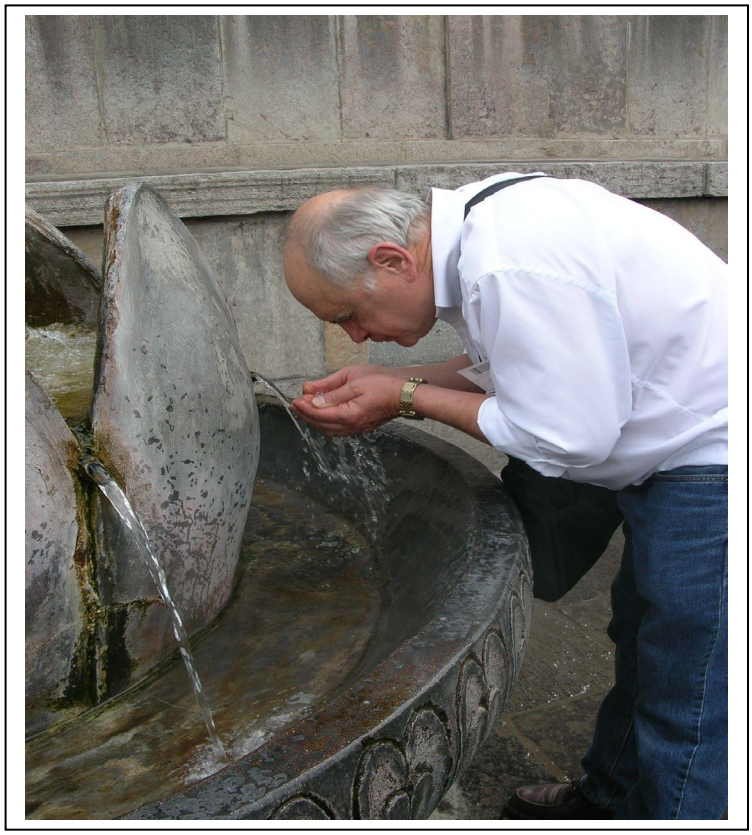
We left Beijing with Dr. He to fly to Xian, home of the famous Terra Cotta soldiers, the "8th wonder of the World." Strange arrangements were made for our stay in Xian; our driver had to come from some distance. His arrival was delayed and so we waited for him at the airport. In Xian, a surprisingly large city, we stayed at the Shangri-La Golden Flower Hotel. The day concluded with a nice dinner at the hotel.

Thursday, 13 September 2007

After breakfast we left with our driver to visit the Terra Cotta Soldiers Museum of the Qin Dynasty about one or so hours from Xian. This famous archaeological discovery was made in 1974. It seems that the Emperor Qin who united China more than 2200 years ago arranged for his afterlife beneath the earth's surface by not only preparing a vast mausoleum, but also prepared an army of terra cotta soldiers, horses, chariots, and weapons including long bows, cross bows, metallic spears, metallic swords, metallic arrowheads, etc. to protect him beneath the earth's surface. This army remained undiscovered for more that 2000 years. Of particular interest was the fact that the sword edges seem to be chrome treated even though the west did not "invent" chrome plating until 1947. The site is unique because the terra cotta artifacts, which number more than 6000 soldiers, have been maintained in their state of discovery, and additional digging continues to discover further information about Emperor Qin's Terra Cotta Army.

Terra Cotta Soldiers Museum of the Qin Dynasty
discovered in 1974, Xian, September 2007

In the afternoon, after lunch, we went with Dr. He and the driver to visit nearby Hua Qing Hot Springs at the foot of Black Horse Mountain. This famous hot springs was visited by emperors and dignitaries to enjoy the hot spring mineral baths. Much of the stone work was black and green jade from Black Horse Mountain. Dr. He purchased a rather large piece of jade, about 4x6 inches, for us after some negotiation with the proprietor of the curio shop. It is a beautiful specimen that will remind us of the good time in the Xian area. Our guide told us many interesting stories about the hot springs, but one of special interest is what is known as the Xian affair when Chiang Kai-sheck, in December 1936, was betrayed and captured by his generals on Black Horse Mountain. Evidently this resulted in the eventual union with Mao Tse-Tung to defeat the Japanese who had invaded China. Finally, after a short rest at the Shangra-La Golden Flower Hotel we took dinner in the center of Xian near the Bell Tower at a famous restaurant that specializes in dumplings. We had 18 different dumplings with an assortment of other goodies before returning to our Golden Flower Hotel to prepare to fly to Guilin on Friday the 14th.



JDM at the Hua Qing Hot Springs at the foot of Black Horse Mountain, September 2007

Hua Qing Hot Springs with Black Horse Mountain
in the background, September 2007

Friday, 14 September 2007

Even though Granny is in pain with her neck and shoulder problems, she has been spectacular during the trip, taking photos, engaging others in conversation and managing all this on a rather limited wardrobe, since our luggage with clothes was missing for several days. She is known as the "Three Pearl, Gold Flower Lady." We are now at the half way point of our trip, during which time we will have been in 6 cities in 12 days with a total travel distance in China of about 5000 km when we complete our journey. It should be mentioned that not only are the accommodations first class, but the domestic flights are frequently first class, such as our flight from Beijing to Xian on Hai Nan Airline.

During the evening in Guilin we celebrated our anniversary by having a boat ride through the Guilin water system, complete with passage through two locks connecting lakes of different elevation. This special surprise was organized by Dr. He as well as a celebration with cake and special music at a night club after the boat tour of the Guilin water system. The party was great with special music and the cake was delicious. The music (piano) was provided by our friend/driver Mr. Mao and his wife.

JDM and Patricia celebrating their anniversary with
a delicious anniversary cake, Guilin,
September 2007

Saturday, 15 September 2007

After an excellent breakfast at the Lijiang Waterfall Hotel-Guilin we departed for our travel on the Li Jiang River through the Guilin Mountains, ~70 km, to Yangshuo. During the boat trip we enjoyed the beautiful Guilin Mountains, water buffalo, rice fields, tea farms, and lovely scenery. For lunch on the boat fresh fish were caught from the Li Jiang River. The water buffalo were especially interesting as they grazed beneath the water for an extraordinary length of time, bringing up mouthfuls of grass, beautiful green color, when they surfaced. In addition to these experiences, it was interesting to see the old fisherman with their cormorants perched on either end of a bamboo pole. These birds are trained to catch and provide fish for their master. Granny and Papa had a photo taken with such a fisherman on the banks of the Li Jiang River. We were accompanied on this boat trip to Yangshuo by Dr. He and Mr. Liang. We were met in Yangshuo by Mr. Mao who had driven our Buick van from Guilin to Yangshuo. After arrival we checked into the Yangshuo Tangrenjie Hotel and then toured the market/stores on West Street. Interesting vendors included a person working with the natural silk cocoons and their preparation/cleaning for eventual spinning into silk cloth. Also, as a bit of a surprise, we

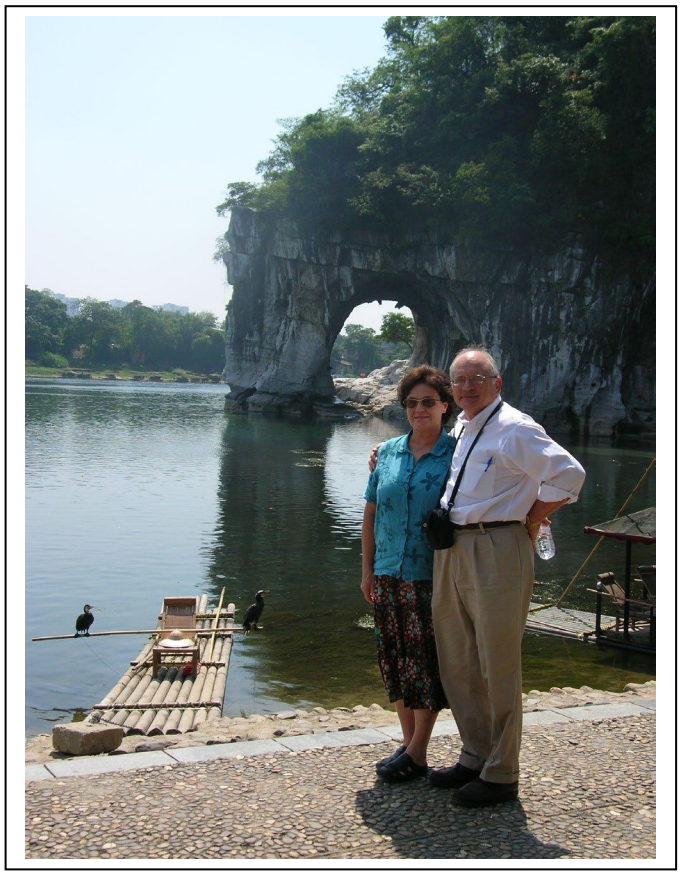
were treated to John Newton's "Amazing Grace" on the flute by a street musician. In the evening we were treated to the spectacular light show, "Impression on ShanjieLiu," on the river surrounded by the Guilin Mountains.

Water buffalo grazing in the Li Jiang River
which flows through the Guilin Mountains,
September 2007

Sunday, 16 September 2007

We began the day by reading the Word, Matthew 5:1-12, the Beatitudes, and prayer to relieve the pain that Granny carries with her. God has truly blessed us on this trip with the opportunity to experience so many interesting aspects of His creation from the geology/beauty of the Guilin Mountains/Li Jiang River, to the marvel of the Great Wall, and to the archaeological discovery of Emperor Qin's incredible Terra Cotta Army in Xian. After breakfast we began the drive back to Guilin with Dr. He, Mr. Lian, and Mr. Mao.

In Guilin, the city of water and mountains, we stayed at the Lijiang Waterfall Hotel-Guilin, with lovely accommodations. The indoor swimming pool was enormous and Papa got to take his first swim of the trip. Upon arrival we first asked for a church and were sent close by to a Catholic Church. The service/mass had concluded but we used their facility to read John 3 to our Chinese friends (He, Lian, and Mao) and tell them a bit about Christianity. During the afternoon we went to the Elephant Hill Park in the city to see a special Guilin Mountain that had formed in the shape of an elephant. At the part we met young girls, representatives of minority ethnic groups including Miao, Dong, Yao, and Zhuan. In the evening, we experienced the waterfall from the top of the hotel (~10 stories) over the complete one side of the building. It was quite spectacular.

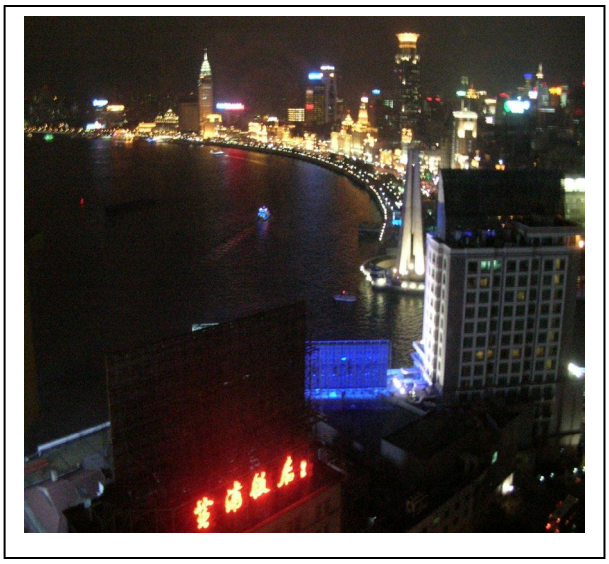


Jan and Patricia at the Li Jiang River – note the Cormorants perched at the ends of the bamboo pole on the fisherman's boat, Guilin, September 2007

Monday, 17 September 2007

We left Guilin with Dr. He, flying first class, to Shanghai where Papa's first lecture was at Donghua University in the afternoon. The lecture, "Advanced Flotation Technology for Removal of Suspended Particles in Wastewater Treatment Plants," went well even though Granny had to leave in the middle of the lecture. Her experience in the bathroom is another story which will be told at another time. The lecture was followed by a tour of campus and a special dinner. The Dean, Dr. Jianshen Liu, presented us with some nice gifts in appreciation of our visit. Finally, we checked into the "Hyatt on the Bund" which just opened in June of this year. This must be a 10 star hotel, the most luxurious accommodations yet, just unbelievable, such opulence. Our room is like a suite having a big picture window overlooking the Bund, the Huangpu River, which runs through the

center of Shanghai. We enjoyed the beautiful night lights from our bed which faces the window.



View from our hotel window, "Hyatt on the Bund," overlooking the Bund, the Huangpu River, Shanghai, September 2007

Tuesday, 18 September 2007

After a fine breakfast buffet, we visited the Oriental Pearl Tower with Dr. He and Dr. Liu Yanan to "view" the city in the rain/fog moving through Shanghai in advance of the typhoon Wipha scheduled to hit Shanghai tonight or tomorrow morning. The 468 meter TV tower is the highest in Asia and the third highest in the world. After lunch at the Shanghai Classic Hotel Restaurant, we visited the 400 year old Chung Huang Miao (Temple). This very nice shopping area was created with architecture from the Ming and Qin Dynasty. In the afternoon Papa had a wonderful swim in the Hyatt's new pool. Similar to all the other Hyatt features, the swimming pool was fabulous. Also in the afternoon we were able to call Ming-yu Wan (6211 0288), Rong-yu's sister and have a nice conversation. We expect to see Ming-yu in SLC next year. Tonight we went with Dr. He to the famous acrobatic performance, Shanghai Acrobatic. This is the same performance that Ginger saw when we were in Shanghai in 1983. It was a great show and after returning to the Hyatt we were prepared to leave tomorrow morning for Changsha. We hope we can avoid the typhoon Wipha, reported to be a super typhoon with winds of 180 km/hr. The typhoon has already caused the evacuation of 2 million people from Fujian and Zhejiang, both south of Shanghai.

Wednesday, 19 September 2007

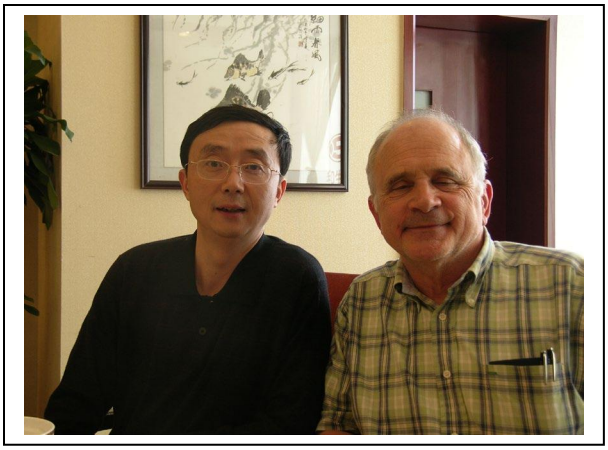
Praise the Lord! Typhoon Wipha was delayed and we were able to escape from Shanghai on a China Southern airline morning flight, flying first class to Changsha. Other travelers were not so fortunate with many delays/cancellations. Wipha is expected to hit Shanghai later in the afternoon today.

Again, the accommodations in Changsha are marvelous. We are staying outside the city in a suite at a hot springs resort, Prees Resort and Hotel. We were met at the airport by Professor Hu Yuehua who had invited us to come to China. In the afternoon we met with Jinshan Li's brother and his friends to discuss water treatment technology. Gifts from Anyang were received including ancient Chinese writing on a bone and a replica of an ancient bronze trough discovered in an archaeological dig. For dinner we went to the old Huo Gong Restaurant in the center of Changsha, a restaurant that Mao Tse-Tung use to frequent when he was a student in Changsha in the 1920s. Out custom prepared traditional Chinese clothes were received tonight, evidently in preparation for the ceremony/seminars tomorrow. These clothes included a wonderful embroidered red silk dress for Granny and a blue silk jacket with dragon for Papa.

Thursday, 20 September 2007

Today we had the special ceremony to appoint Papa as Honorary Professor at Central South University. The ceremony was conducted by Dean, Professor Hu Yuehua, attended by more than 100 student and faculty, and the prestigious appointment was made by Vice President, Professor Guanzhou (Cho). Granny accompanied Papa and both were dressed in the traditional Chinese embroidered silk clothes, the red silk dress for Granny and the royal blue silk jacket for Papa. After a review of accomplishments and collaboration with Central South University for almost three decades, the appointment was made. After that, Papa made remarks of appreciation at the ceremony regarding the hospitality of Central South University and the longstanding relationship with his research group at the University of Utah.

JDM and Patricia at the entrance to Central South University, Changsha, September 2007



Dean, Professor Hu Yuehua, Central South University, and JDM, Changsha, September 2007

JDM and Patricia in traditional Chinese embroidered silk clothing in preparation for ceremony, Changsha, September 2007

JDM with VP, Professor Guanzhou Qiu, being appointed Honorary Professor at Central South University, Changsha, September 2007

Dean, Professor Hu Yuehua, VP, Professor Guanzhou Qiu, Patricia, JDM, and Chinese colleagues at the presentation ceremony, Central South University, Changsha, September 2007

The ceremony was followed by technical lectures, one in the morning, "Molecular Dynamics Simulation for the Analysis of Interfacial Chemistry in Naturally Hydrophobic Mineral Systems," and one in the afternoon, "The 3D Analysis of Multiphase Particles and Particle Beds by Cone Beam X-Ray Microtomography for improved Understanding of Mineral Processing Systems." After the initial ceremony, Granny went shopping and was quite successful, finding pearls of great price. No wonder she is called the Three Pearl, Golden Flower Lady. In the afternoon, Granny went on a hike to visit some gardens at a mountain adjacent to Yelou Mountain where Chairman Mao would go for meditation and writing. The day concluded with a nice dinner at the Prees Resort and Hotel and Restaurant where we stayed while in Changsha.

Friday, 21 September 2007

Granny packed our luggage Thursday night in preparation for our first class flight to Beijing today and on to the U.S. tomorrow. This was a great undertaking since we have acquired many gifts, so much so that we had to purchase a new suitcase. It was amazing to see how efficient Granny was in getting this job done. Hard to believe that all our treasures were accommodated.

During the morning Granny met with Zhong Hui, Dr. He Zhiguo's wife, and had an interesting discussion about religion and Christianity, in particular. At the same time Papa toured the Central South University laboratories and discussed possible research collaboration regarding bauxite ore/clay mineral surface chemistry, molecular dynamics simulation, and iron ore pelletization research using the Funa binder (already CSU has had or has projects with CSN and CVRD from Brazil).

For lunch we went with Dr. He, Dr. We Sun, and Professor Li to the famous Xihu Lou Restaurant, listed in the Guiness Book of Records for being the largest restaurant in Asia, and perhaps the world, being able to serve 5000 patrons at the same time. After seeing some live stage, entertainment, traditional Chinese dancing, we were still able to reach the airport to catch our afternoon flight to Beijing on China Southern Airline. During the flight first class service included the distribution of flowers, carnations. After checking in at the Xiyuan Hotel and having a bit of a rest, we were invited to have dinner with principals of Kongnoy Investment Company regarding their interest in establishing a joint venture water treatment company in China.

Saturday, 22 September 2007

After breakfast we departed with Dr. He for the Beijing International Airport to catch our United flight to San Francisco. Unfortunately we were not able to meet Tao Jiang before leaving. Tao Jiang was coming to meet us at the airport to discuss Funa pelletization research. Before departure we were able to purchase a handmade vase of beautiful color and texture. The return flight of about 11 hours to the U.S. was fine with seats on the exit row. We were able to take an earlier than scheduled flight from San Francisco to Salt Lake and arrived exhausted at about 1 p.m. What a wonderful trip. The change in China has been incredible. We saw so much in 12 days (6 cities and 5000 km) bringing back many past memories from our trip with Ginger and Ron-Yu in 1983 and current memories/treasures from our trip in 2007 thanks to Hu Yuehua, He Zhivguo, and many Chinese friends.

Journal of Trip to Yunnan Province
P.R. of China
J.D. Miller
16 – 29 May 2009

Saturday, 16 May 2009

Departed at about 8 am from Salt Lake with Xuming Wang on United 837 to San Francisco. We arrived without difficulty and began a 6 hour wait for our Air China 986 flight to Beijing. We had a breakfast while waiting and worked on two papers, one dealing with the fundamentals of interfacial water and the structure of adsorbed dodecyl amine at a silica surface and the other paper dealing with flotation technology for the recovery of trona. Finally we departed at about 3 pm with business class accommodations.

The trip to Yunnan Province in China had been organized by Xuming and Qiang Yu at the request of the Yunnan Department of Science and Technology, and the China Yunnan Metallurgy Group (CYMG), now CYMCO. In addition to visits to operations in Yunnan, our schedule included meetings with the faculty/students at Kunming University of Science and Technology (KUST), with the Yunnan Department of Science and Technology and with CYMCO, as well as with other industry representatives. CYMG had arranged for me to fly business class which made the trip much more comfortable. Yunnan has an abundance of mineral and metal resources, being the number one tin producer in the world. This region of China is known as the "High Country." The elevation at Kunming is about 6800 feet. The climate is quite comfortable and Kunming is known as the Spring City. It has four seasons; spring, spring, spring, and spring.

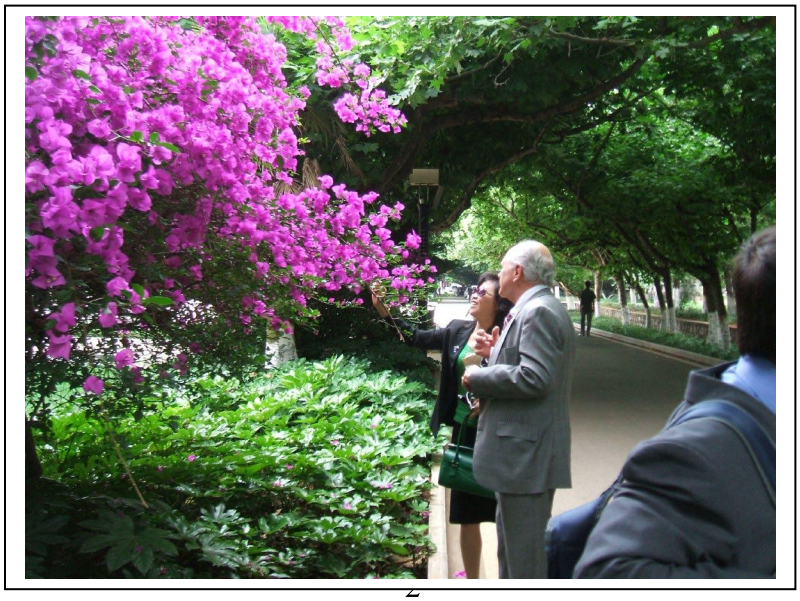
The flight to Bejing was about 12 hours and, after a 3 hour wait in Beijing, we flew on to Kunming in Yunnan, arriving at about 1 am Monday morning, of course losing a day when we crossed the International Date Line. The total travel time was about 26-27 hours. Needless to say we were exhausted. We were met in Kunming by Qun Zhao, Director of Technolgy for CYMCO Research Institute and his entourage. A great bouquet of lilies as well as other floral arrangements celebrated our arrival. Finally we made it to Kunming Hotel and to bed by 2 am. Wei Xia from CYMCO, who later I discovered professed to be a Christian, served as translator and organized business cards for me since I forgot to bring cards with me.

Monday, 18 May 2009

After breakfast we spent the morning at the Yunnan Department of Technology and Science discussing the status of mineral processing/metallurgy technology in Yunnan. At this meeting I received my appointment as Consultant on Science and Technology Development Strategy from Director-General Long Jiang. After lunch at Green Lake Restaurant, we went to the university to prepare for my lecture.

The campus at Kunming University of Science and Technology is beautiful with unique trees (bougainvillea) having leaves with a beautiful purple color. Vice President Jinhui Peng introduced me for the lecture. See appendices for Professor Peng's introduction. Many of the faculty from Metallurgy and Mineral Processing along with several hundred students attended the lecture, "Mineral Processing and Hydrometallurgy Research." We had some good discussion after the lecture with one of the students who had misunderstood my comments about Edwin Abbott's "Flatland" asking about the Bible. In this way I had
opportunity to highly recommend that he and others read the scriptures. This was in the presence of the faculty/student audience. Of course the event was well advertised with posters having my photograph which they had taken from our website.

JDM, Qiang Yu and Xuming Wang, Kunming University of Science and Technology, Yunnan, May 2009



JDM with Wei Xia admiring the beautiful trees (bougainvillea), Kunming University of Science and Technology, Yunnan, May 2009

After the lecture, which took a couple hours, we toured laboratories before going for dinner with the President of the University, Professor Zhou Rong. Other faculty to mention included Jinhui Peng, Vice President, Wenbin Zhang, mineral processing faculty, who had some good comments and discussion after the lecture and was Keqing Fa's professor at Kunming, Deng Gang (Gary), Professor and Director of International Cooperation, and others, including one professor who had been at the U of U for 6 months last year. The President and faculty seemed to be keen to establish international collaboration and we discussed the possibility for remote participation in the MODSIM short course in June.

Tuesday, 19 May 2009

It should be mentioned that many former students/researchers graduated from Kunming University of Science and Technology (KUST) and include,

Ye Yi
Qiang Yu
Xuming Wang
Minhua Li
Jinshan Nie
Keqing Fa
Yin Di
Ning Liu

During the visit at KUST on Monday I was able to meet some of their former professors and after the lecture I was presented with an artisan pewter vase, most appropriate because Yunnan is known for their nonferrous metals industry particularly tin.

After breakfast we went to CYMCO Research Institute (CYMCO is now only 70% owned by government) still in Kunming to meet with the research and technology staff and give a lecture, "Developments in Copper Ore Processing." After the lecture we had several hours of discussion before lunch. Qun Zhao, Director of Technology, and several researchers, Haiphang among others participated in the discussion.

After lunch, we had a rest at the CYMCO's hotel and then met with Dong Ying, Chairman of the Board, and a dozen of his vice presidents for different divisions, mining, metallurgy, aluminum operations, lead/zinc operations, as well as titanium and steel. Representatives from Yunnan Copper Company and Yuntainhua Company (phosphate, potash, kaolinite) participated in this meeting. CYMCO employs 22,000 with most of their operations in Yunnan.

Much of their energy comes from hydroelectric power. Chairman Dong Ying is a classmate of Qiang Yu and Xuming Wang and is keen to continue collaboration in research/technology at the University of Utah. He was especially pleased that I had been appointed consultant by the Yunnan Department of Science and Technology. The meeting concluded in Dong’s office where he gave us some of the famous red tea from

Yunnan. After further discussion we went for dinner at a rather luxurious Fortune Restaurant where we were greeted in a chant or salutation by a chorus of young ladies.

Of course I was exhausted by the end of the day. It will take some time to adjust my bio-rhythm. I was still awake at 3 am.

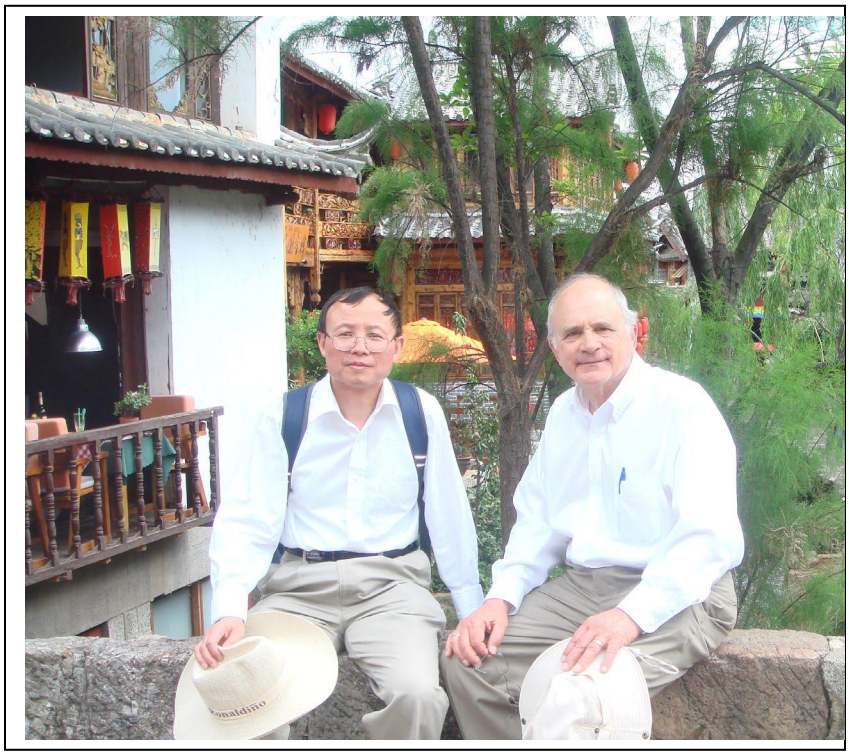
Wednesday, 20 May 2009

This morning after breakfast we leave Kunming to journey to Dali (Xuming's hometown) and eventually on to Lijiang, still in the province of Yunnan. We will travel most of the day in a brand new GM van called the Presidential. This van has everything, TVs, telephones, and even seats with an automatic massage capability. Patricia this would serve you well. We need to have such for our next car. Now you should know that Yunnan is in the south west corner of China and borders with Viet Nam, Cambodia, Laos, India, and Tibet.

The trip west from Kunming to Chuxiong (chew shawn), Dali (Xuming's hometown), and finally Lijiang took all day in the GM Presidential van. Leaving at 8:30 am, we had Wei Xia, Dr. Peng, Guo Hong, Xuming Wang, Qiang Yu, and myself. The first stop, as we made our way toward the Himalaya foothills, was Chuxiong, home of the Yi ethnic people and the land of dinosaurs. Homes in this Chuxiong district were painted white and each home was decorated with a large circular painting, either a geometric design or an animal, as I recall. In some regard, it reminded me of the Amish country. A nice tour of the museum was organized after which we had lunch before proceeding to Dali. Xuming departed from us at Dali and we went on to Lijiang, quite a tourist center with millions coming to visit each year. As we approached Lijiang, the great mountain, the Yulong glacier, could be seen in the sunset. We had dinner and then checked in at the Lijiang Jian Nan Chun Hotel.

Thursday, 21 May 2009

Well here we are in Lijiang looking at Yulong, at the beginning of the Himalayas, ~200km from Tibet. The elevation at Lijiang is about 2500 meters. Today we toured the old town and the old village, Suhe, which was destroyed by an earthquake in February 1996 with 10,000 fatalities. In years gone by this location at Lijiang was a center, a horse station, to carry trade from east to west through Tibet. The morning concluded with a tour of the museum including riding of the horses around the old village.



Qiang Yu and JDM, Lijiang Village (2500 meters in elevation),
Yunnan, May 2009

The afternoon centered around a swim at a hot springs resort, the Sun, Moon Reflection Lake Spring. The temperature was about 90 degrees. Nevertheless I was able to swim 10 laps. A race with Guo Hong was also included. Guo Hong is a good swimmer. I know you would have enjoyed the foot massage afterward.

In the evening after dinner we attended a folk music concert performed to represent the Naxi ethnic group from this region.

Friday, 22 May 2009

Still in Lijiang. It should be mentioned that we changed hotels, moving to the 5-Star Guanfang Hotel on Tuesday. This move was quite an improvement and we are enjoying the current accommodations.

This morning after breakfast, we traveled to the Yulong Glacier or the Jade Dragon Glacier (~5500 meters elevation). This region is the beginning of the Himalayas, not far from Tibet. We took a cable car for most of the trip but had to climb the final 1000 meters, reaching about 4600 meters. The expedition certainly was a special way to experience more of God's creation.

JDM and Qiang Yu at the Yulong Glacier, Yunnan, May 2009

In the afternoon we saw the live production of "Impression Lijiang" in an amphitheater. The production describes the culture of the Naxi ethnic group with hundreds of participants and is directed by the person who organized the Beijing Olympics. The Yulong Glacier provides the backdrop for the theater. The story not only involves the Naxi culture but also features the horse station at Lijiang used for trade along the ChaMa ancient road into Tibet. The breed of horses was critical to the success of caravans traveling from east to west, and from west to east. These horses have short powerful legs able to manage heavy loads on the rugged mountain ChaMa road, only a few inches wide at some points.

Live production of "Impression Lijiang" in an amphitheater describing the culture of the Naxi ethnic group and directed by the person who organized the Beijing Olympics, Yunnan, May 2009

The horses were a big part of the Naxi culture. The Naxi's were well known for their horsemanship. They were also well known for their custom that the husbands should not be burdened with household matters. Women were expected to do the housework, farm the fields, bring up children, and be proud of hardworking.

In the evening we went to the National Culture Exchange Center Theater for some folk dancing, followed by dinner.

Saturday, 23 May 2009

The last three days (Thursday through Saturday) in Lijiang have included Qiang Yu, Guo Hong, Jinhui Peng, and Wei Xia who provided an extraordinary program of activities which I have tried to describe.

After an excellent breakfast at Guanfang Hotel where the manager, Antonio....., searched all over Lijiang to find me decaffinated coffee, we began the day with a tour of He Long Tang (Black Dragon Lake). This lake is fed by nine glacial springs. We then traveled north about 15 km just close to the Yulong cable car to visit Yu Shui Zhai (Jade Water Village). Here the Naxi culture was further described including their God of Nature and the sacrifices that were made. No explanation was given regarding the purpose of the sacrifice or of life after death. The tour included a Yak ride and preparation of salmon for our lunch. The preparation first required that a salmon be caught from the glacial spring pond. In this regard Jan landed a > 2 kg salmon which was then used for the following luncheon dishes; Sushi, Head and Tail Bone soup, Fried Skin, and what appeared to be Poached Salmon.

JDM and Qiang Yu with some Chinese colleagues riding horses in Lijiang, Yunnan, May 2009

JDM and his yak friend, Yu Shui Zhai (Jade Water Village), Yunnan, May 2009

The afternoon concluded with some shopping in Lijiang Old Town where we organized Yulong (Jade Dragon) Glacier Mountain tee shirts for the grandchildren. This required some custom design since they did not have exactly what I wanted. Finally we were able to locate and purchase a rock containing Jade from the Yulong Glacier Mountain.

In the evening we were invited for dinner by Antonio..... at his five-star Hotel in the city. A five course dinner was given with excellent presentation. Special Kobe steaks from Japan were served as the main course. Everyone agreed the steaks were delicious. After dinner I had a swim in the Hotel's pool. It was a full day, but quite enjoyable.

Sunday, 24 May 2009

Some further update on the previous journal entry for Saturday 23 May 2009 would be in order. First, we were fortunate to ascend Yulong Glacier on Friday because on Saturday the weather was not quite as nice. In fact there was a storm on the mountain with snow fall. The second story of interest is that while in the shop organizing the tee shirts and having some custom work done, a small Chinese child, perhaps the son of the proprietor, about 2 years old, came to me to say "hello" and called me "Yeye" or grandpa. He then sat with me on my lap. He is a cutie and I know Granny would love him.

Qiang Yu and JDM with young Chinese boy in Lijiang shop who came to greet me, calling me “Yeye,” Yunnan, May 2009

Today we left Lijiang after breakfast, driving back to Kunming in the Presidential van. As mentioned early-on, the trip is an all-day drive. Everything went well. We stopped in Dali to pick-up Xuming and had the opportunity to meet his father, mother, and brother. Everyone was really excited to see us. Xuming has a very nice family.

JDM with Xuming Wang, his father (Jianjung Wang),
mother (Wen Qun Zhang), and brother (Xuguan Wang),
Dali, Yunnan, May 2009

We traveled on to the Yunnan Xiang Yun Feilong Nonferrous Metalco, Ltd. zinc plant for a tour and discussion with Yang Long, General Manager, and his staff. Suggestions were made regarding the concentration of low grade feed containing the zinc carbonate mineral, smithsonite, I believe. In addition process strategies eliminate the thickening/filtration currently being used. These strategies included adsorption from the leach slurry and heap leaching not only to eliminate current thickening/filtration operations but also to eliminate the grinding circuit. Yang Long presented us with a silver bracelet made from the silver metal produced from their operations.

After our visit to the zinc plant we returned to Kunming at about 7 pm and had dinner at a Korean restaurant. I sure have had some strange food on this trip, not unexpected. On this occasion the food caused some discomfort for about one day. Nevertheless, I've survived so far with advice from Qiang Yu and Xuming Wang.

Monday, 25 May 2009

Now we are back in Kunming. The Yuntianhua Group was visited to discuss phosphate processing technology. The phosphate ore extraction division ranks number two in the world in phosphate production from 5 major mines. The company has great laboratories for mineral processing, world class. The company's research institute attends to special flotation reagent preparation. However, based on our discussion, greater attention should be given to particle characterization in order to improve process strategies. The Yuntianhua Company seemed to be interested in collaboration with the University of Utah and possible projects are being considered for collaboration.

After lunch Xuming and I visited at the Kunming University of Science and Technology for discussions with the Mineral Processing Faculty. The university has a long tradition of mineral processing dating from 1925. Among other matters discussed, plans for participation in our MODSIM Short Course were made.

The day concluded with dinner at another campus including Wei Xia, Ma, and Zhang from CYMCO, Guo Hong, Li, and Wang from Yunnan Department of Science and Technology, and Vice President Jinhui Peng from Kunming University of Science and Technology. Many of these individuals (Ma, Zhang, Li, and Wang) had visited the U of U in November 2008.

Tuesday, 26 May 2009

We traveled four hours south from Kunming to Gejiu in the Presidential Van, not too far now from the southern borders with Viet Nam, Laos, and Myanmar (Burma). The climate is warm and more humid. This is the great region of China where world tin resources have been supplied at least since the Han Dynasty, and, who knows, maybe back to the bronze age.

The Yunnan Tin Company (YTC) in Gejiu, Yunnan, was visited to discuss mineral/metal technology with Chen Rui Han, Vice General Manager, and his staff and to tour the smelter/refinery. YTC is the world's largest tin producer with 8 mines and 4 mineral processing plants in operation. Both oxide and sulfide ores are mined (~6M tpy) and processed by gravity in the case of oxide ore and by flotation and gravity in the case of sulfide ore. The larger plants process about 2000 tpd. In addition to refined tin, a copper concentrate and a lead concentrate are made. The feed grade is generally about 0.7% and the material is ground to minus 0.5 mm. The average recovery is ~66%. A review of the mineral processing program at the University of Utah was presented by Xuming Wang as well as new technology including X-ray micro CT for 3D analysis of multiphase systems and air sparged hydrocyclone technology. A need was expressed by YTC for the development of new process technology to treat fine feed and for the development of new tin products. Certainly more detailed process mineralogy would help improve the YTC operations.

Chen Rui Han presented us with a sample of the sulfide ore of tin and a gift of a tin container for tea during dinner at the Yunxi Hotel. We stayed at the five star hotel, the Guanfang Hotel, for the night. Unfortunately the air conditioning was not working. It was interesting to visit and experience this historic and still active mining region for tin production. As mentioned previously Yunnan is the world's largest producer of tin.

Wednesday, 27 May 2009

I had breakfast at the Guanfang Hotel in Mengzi, a town close to Gejiu and then went for a swim in the pool while Xuming Wang, Qiang Yu, Wei Xia, and Guo Hong went to have special Chinese noodles for breakfast, Xuming's favorite dish for breakfast. We have a photo showing the immense bowl of noodles Xuming had for breakfast. We then departed for the Mengzi Mining and Metallurgy Company with Li Yu, Chairman, and his staff.

Xuming eating his special noodles for breakfast, Mengzi, Yunnan, May 2009

Operations at the Mengzi Mining and Metallurgy Company were toured and hosted by Li Yu, Chairman of the Board. A polymetallic ore is mined containing lead, zinc, tin, copper, etc. Engineers and staff showed us the plant operations for zinc and indium production. Some former students from Central South whom Patricia and I met in 2007 were at the Mengzi operation. They were very happy to see me again and we had good conversation about technical matters. Indium production at Mengzi is one of the highest in China. Products such as lead and tin concentrates are shipped to other plants for

processing. The zinc concentrate (600ppm indium) is dead roasted at 850 degrees centigrade in a fluid bed roaster, the calcine being leached with sulfuric acid. The leach solution is purified before electrolysis to recover zinc/indium in the cathode. The cathode is melted and then fumed to recover indium in the dust. The dust is leached in acid sulfate and separated by SX using TBP. After stripping with hydrochloric acid the indium is recovered by cementation on zinc plates, cast into anodes and refined electrolytically. After the plant tour, dedication to science and technology was discussed and technology achievements in cooperation with Central South University. The company was particularly interested in new product development for the indium market in addition to advanced processing technology and environmental issues. Chairman Li Yu presented me with an indium ingot (99.99%) and a beautiful hand crafted vase from local artists. I received so many wonderful gifts that a new piece of luggage must be purchased to manage my return to the USA.

JDM touring the Mengzi Mining and Metallurgy Company,
where zinc, indium, and other metal products are produced,
Mengzi,Yunnan, May 2009

Chairman Li Yu presenting JDM with an indium ingot (99.99%) and a beautiful handcrafted vase from local artists, Mengzi Mining and Metallurgy Company, Yunnan, May 2009

After lunch we began the four hour return to Kunming and ended the day by having dinner with researchers who had visited for 6 months at the University of Utah for the Bauxite project. We reminisced about their experience in Utah before retiring for the day. They were very pleased to meet with us.

Thursday, 28 May 2009

Today we went to the Stone Forest, World's Heritage List by UNESCO. The Stone Forest is about 1-2 hours east of Kunming and so we were able to complete the tour in the morning and return to Kunming for lunch. This vast limestone forest of stone (the park is 20 square kilometers, I believe) seems to have formed from a marine environment and has experienced extensive weathering giving rise to thousands of "Trees," maybe hundreds of thousands. The park is open 24 hours and of course at night everything is lighted. The ethnic Yi people live in this region and provide ethnic color to the park. In addition to the vast stone forest with "trees" reaching 50-100 meters in height, I suppose, the park has some rice fields maintained by the Yi people and various plants, Bougainvilla, for example, and birds, the Bugu who seemed to call to us as we visited and walked around the park.

JDM and two ethnic Yi people with the Stone Forest in the background, Yunnan, May 2009

JDM in the Stone Forest with a Yi friend, Yunnan, May 2009

The day was hosted by Wei Xia as Qiang and Xuming had business meetings. Wie Xia organized my visit and has been the translator for our visit. She has been with us for all of the two weeks. It was exciting to learn that Wie Xia became a Christian about five years ago and so after a delightful lunch at a very historic restaurant, we went to see the church she attends. This is the Trinity International Church which origin goes back to 1903 when it initially served as the mission station for the China Inland Mission. Recently in 2003, they broke ground for the new church building which we visited today. So the church is over 100 years old and it is estimated that they baptize 500 individuals each year. See appendices for further information. I promised to send Wie Xia the book of John for her to begin study of the bible.

Trinity International Church, founded in
1903, which Wei Xia attends

Now we must attend to packing and prepare for the trip home on Friday. It has been an incredible excursion. Professor Peng, Vice President of KUST, presented me with a unique gift, a specimen of Burma Tree Fossil Jade. See appendices for further information. This beautiful jade is a nice addition to my collection of Chinese jade. I'm sure I have missed some of the interesting/exciting moments. I'm saturated with Chinese food but am rejoicing in the Lord for His goodness.

Friday, 29 May 2009

I completed my report (DST) and submitted it electronically to Guo Hong (DST) last night, Thursday evening. See copy of report in appendices. Thursday was a trying day for Qiang and Xuming regarding authorization to proceed with our bauxite project sponsored by CYMCO. It seems that the Institute Director, Qun Xhao, has been inconsistent and is

difficult to trust. Nevertheless the issues are being resolved and we will proceed with the 2nd phase of the bauxite project with new researchers coming in June. Also, Wie Xia brought the new luggage last night to pack all my acquisitions. Packing was completed and we (Qiang Yu, Xuming Wang and I) departed for the airport at about 8:30 am accompanied by Guo Hong (DST). Wie Xia and Qun Zhao, both of CYMCO, said goodbye at the hotel. We had 1st class check-in at the airport where we were met by Long Jiang (Director General, DST).

The flight to Beijing was fine and there we were met by old students and research visitors, including Fangqin Cheng, her student, Hao Du, and Zheng Shili. Dr. Cheng wanted me to come to Shangxi and I told her that I would try to arrange something next year. We had no difficulty with check-in although my luggage was overweight. Praise the Lord who was so gracious to me on the trip and provided in all cases even in the issues that were discussed last night with Qun Xhao.

Shili, Hao Du, JDM, Fangqin Cheng, Xiaoli, and Xuming at the Beijing International Airport

The flight to San Francisco went well. The business class was not full, lots of room, and I had a good sleep. Custom clearance and recheck of baggage was without difficulty and the flight to Salt Lake City was as scheduled. I had called Patricia and she collected me upon arrival at about 7:00 pm in SLC.

APPENDICES

Welcome Speech by KUST Vice-President Prof. PENG Jinhui
for the Visit of Prof. Jan D. Miller from the of University of Utah, USA
May 18, 2009, KUST Lianhua Campus, Kunming, Yunnan China

Good Afternoon, your excellency Professor Jan D. Miller,
Director GUO Hong from Yunnan Provincial Department of Science and Technology,
professors and students, ladies and gentlemen.

It is a great honor for me to be here today, on behalf of Kunming University of Science and Technology (KUST), to welcome the visit of Prof. Jan D. Miller to this university.

I had the pleasure of visiting the esteemed University of Utah in the USA last year, and learned some of the great achievements made by Prof. Miller and his teams in metallurgy and other fields. As a university in China with one of the oldest and strongest department of metallurgical engineering, KUST has been looking forward to the chance of learning form Prof. Miller and welcoming your visit to our campus. I am especially happy personally today, because I used to be a student and then the Dean of the Faculty of Materials and Metallurgical Engineering of this university.

Historically, KUST started in 1954 as Kunming Institute of Technology, with the main focus, at that time, only on the mineral industry of the south west region China. Today, the relevant degree programs and research fields of mineral industry are still the strongest fields of KUST. Of course we have had quite some development in addition to the traditional fields. Now KUST has the Faculties of Arts, Law, Science, Social Science, Management and Economics, etc. besides many Faculties of engineering, and the student population is over 50,000, including 6,000 graduate students in Master and Ph D programs. KUST is currently offering nearly 250 degree programs including Bachelor, Master, and Ph D degrees. With these data, in fact KUST has become the largest university in Yunnan province, a province with 45 million people. I believe that KUST and the University of Utah both have good potential to cooperate with each other, of course first through the fields of metallurgy and the relevant areas. I understand that metallurgy in the USA nowadays is a little bit not as active as many years ago. But in China, it is still an active field due to the development of the relevant industries. KUST has very close relationship with many large metallurgical enterprises like Yunnan Metallurgy Group, and we are also helping the neighboring south-east Asia countries like Vietnam to develop their higher education in metallurgy. Therefore, your rich knowledge and experiences in metallurgy will be very helpful and valuable to us, and to the sustainable development of metallurgical industries in this region of the world. From the viewpoint of your side, I believe you may also be much interested in knowing the situation and getting involved in the academic and industrial development of metallurgy in this region. I do hope through your visit and our later-on communication, we will find common interests such that sound cooperation can be established between us, which shall be creating win-win situation to both sides. For today, I believe the students and staff of KUST will benefit from your lecture first, for which we appreciate very much. And I also hope that you will be enjoying the visit and communication here. May I take this opportunity, to wish you a very pleasant trip in China and back home?

Thank you very much.

Visiting Program for Prof. Jan D. Miller

at Kunming University of Science and Technology (KUST)

May 18, 2009

Time	Activity	Note
14:30 — 16:30	**Brief welcome session followed by seminar** Chair: Prof. DENG Gang, Director of International Cooperation, KUST **1. Welcome address by Prof. Dr. Jinhui PENG, KUST Vice President** **2. Seminar delivered by Prof. Jan D. Miller** **Topic:** ***Research in Mineral Processing and Hydrometallurgy*** **Venue:** Lecture room, Library, Lianhua Campus, KUST **Participants:** Division of International Cooperation Division of Research Administration School of Graduate Studies Faculty of Land Resource Engineering Faculty of Materials and Metallurgical Engineering Faculty of Mechanical and Electrical Engineering Faculty of Environmental Science and Engineering Key Laboratory of Fabrication and Processing of Advanced Materials Office of General Affairs	
16:30 — 18:00	**Visiting labs / Research Institutes, Lianhua Campus, KUST** -Yunnan Provincial Research Institute of Energy-saving and New Technology for Energy -National Engineering Laboratory of Nonferrous Metal and Vacuum Metallurgy -Geological Relics Research Center of Ministry of Education -Research Institute of Yunnan Provincial University/College Material Processing and Engineering Material -Key Laboratory of Unconventional Metallurgy of Yunnan Provincial University/College **Participants:** Prof. Dr. Jinhui PENG, KUST Vice President Division of International Cooperation Division of Research Administration School of Graduate Studies Office of General Affairs	
18:30 — 20:30	**Welcome dinner hosted by Prof. Dr. Jinhui PENG, KUST Vice President** **Venue:** Teacher's canteen at Lianhua Campus **Participants:** Office of General Affairs Division of International Cooperation Division of Research Administration School of Graduate Studies, Faculty of Land Resource Engineering Faculty of Materials and Metallurgical Engineering Faculty of Mechanical and Electrical Engineering Faculty of Environmental Science and Engineering Key Laboratory of Fabrication and Processing of Advanced Materials	

Report on Yunnan Visit, 16-29 May 2009
Guo Hong
Director, Division of Science and Technology Yunnan Province, China

GENERAL INTRODUCTION AND INFORMATION

Initial meetings were held with Yunnan Provincial Science & Technology Department, Kunming University of Science and Technology-Faculty from Metallurgy and Mineral Processing, and with CYMCO representing the mineral/metal (Lead, Zinc, Aluminum, Silicon, Ti, and Metal Alloy) industry in Yunnan to obtain an appreciation of the technology status of the mineral/metal industry in Yunnan province. Recognition must be given to Guo Hong, Director of Division of Science & Technology Cooperation Between Province and Chinese Academy of Sciences/University, Yunnan Provincial Science & Technology Department, and Long Jiang, Director General, Yunnan Provincial Science & Technology Department, Dong Ying, Chairman of the Board, CYMCO; Zang Jian, Director of CYMCO's Science & Technology, and Jinhui Peng, Vice President Kunming University of Science and Technology; all who made this visit possible. My best students and excellent friends Dr. Qiang Yu and Dr. Xuming Wang have dedicated their time and energy to accompany me on all activities in Yunnan during this visit. Without their assistance, I would not have had such a great enjoyable trip. Also recognition is given to Wei Xia of CYMCO for her organization of all details.

In the meeting at Yunnan Provincial Science & Technology Department with Long Jiang, Director-General, Department of Science and Technology, Mr. Luo Guo Quan, Deputy Director General, Guo Hong, Director, Division of Science and Technology, and other staff, the mineral/metal production in Yunnan was discussed and the three-year appointment as Consultant on Science and Technology Development Strategy was made. Yunnan Provincial Science & Technology Department's future strategic development was reviewed by Director-General Long Jiang, Yunnan is second largest province in the mineral production in China, including Tin, Lead/Zinc, Copper, Aluminum, Coal, Phosphate, Bauxite. In the case of tin, Yunnan is the number one producer in the world.

The first meeting at Kunming University of Science and Technology, hosted by Vice President Jinhui Peng, was quite successful. Many of the faculty from Metallurgy and Mineral Processing along with several hundred students attended the lecture," Mineral Processing and Hydrometallurgy Research". We had some good discussion and tour of the laboratories after the lecture. The next week we met with the Mineral Processing Faculty and were briefed on their activities as well as discussing collaboration in education and research. Some areas for collaboration included participation in mineral processing plant simulation short-course. It seems Kunming University of Science and Technology has made significant progress on research projects and can be prepared to be competitive at the national level but better efforts must be made to compete with Central South University which has done $8M worth of research in Yunnan. Development of areas of expertise to be recognized by government and industry is necessary such as microwave

technology that has been developed in Metallurgy by Professor Peng. For example, the Mineral Processing Faculty could develop expertise in simulation and/or mineral liberation. Already the decision to participate in the University of Utah short course on plant simulation is a step in the right direction.

The meeting with industry representatives was organized by Guo Hong and hosted by Dong Ying, Chairman of the Board of CYMCO and held at CYMCO's offices. A number of CYMCO's vice presidents for different divisions, mining, metallurgy, aluminum operations, lead/zinc operations, titanium and steel, as well as representatives from Yunnan Copper Company and Yuntianhua (Phosphate Chemical Group). Based on this meeting a number of research needs by CYMCO were identified including projects involving bauxite, titanium, and others. A detailed discussion has been made with CYMCO on the CYMCO sponsored bauxite flotation project as an example on technology development (now in progress). The first half year of the bauxite project has been completed. As a result, not only has progress been made and research for the second half of the project will be continued but a paper has been accepted for publication in an international journal.

COMPANY MEETINGS AND PLANT TOURS

Meeting with the CYMCO’s Research Institute staff was held to discuss CYMCO technology and research needs. CYMCO employs 22,000 with most of their operations in Yunnan. Much of their energy comes from hydroelectric power. Prior to the meeting with industry, a lecture, "Developments in Copper Ore Processing" was given at the CYMCO Institute followed by several hours of discussion with Qun Zhao, Director of Technology for the Institute and several researchers participated in the discussion as well.

We traveled to the Yunnan Xiang Yun Feilong Nonferrous Metalco, Ltd. zinc plant for a tour and discussion with Yang Long, General Manager and his staff. Suggestions were made regarding the concentration of low grade feed containing the zinc carbonate minerals, smithsonite I believe. A suggestion to use raw feed material to calculate process efficiency was made to Yang Long during discussion. In addition process strategies eliminate the thickening/filtration currently being used. These strategies included adsorption from the leach slurry and heap leaching not only to eliminate current thickening/filtration operations but also to eliminate the grinding circuit.

The Yuntianhua Group was visited in Kunming to discuss phosphate processing technology. Discussions/presentations were made with Zhu Mingsong Deputy General Manager; Liao Xiao Jun, Minister of Science and Technology; and staff from both operations and research institute. The University of Utah’s phosphate, soluble salts, and Air-Sparged Technology were reviewed during our meeting. The Yuntainhua phosphate ore extraction division ranks number two in the world in phosphate production from 5 major mines. The company has great laboratories for mineral processing, world class. The company's research institute attends to special flotation reagent preparation. However, based on our discussion, greater attention should be given to particle characterization in order to improve process strategies. After

reviewing University of Utah research program with CYMCO, Yuntianhua company seemed to be also interested in collaboration with the University of Utah and possible projects are being considered for collaboration.

The Yunnan Tin Company (YTC) in Gejiu, Yunnan was visited to discuss mineral/metal technology and to tour the smelter/refinery. YTC is the world's largest tin producer with 8 mines and 4 mineral processing plants in operation. Both oxide and sulfide ores are mined (~6M tpy) and processed by gravity in the case of oxide ore and by flotation and gravity in the case of sulfide ore. The larger plants process more than 2000 tpd. In addition to refined tin, a copper concentrate and a lead concentrate are made. The feed grade is generally about 0.7% and the material is ground to minus 0.5 mm. The average recovery is only ~66% and 72% at the best. A review of the mineral processing program at the University of Utah was presented by Xuming Wang as well as new technology including X-ray micro CT for 3D analysis of multiphase systems and air sparged hydrocyclone technology. A need was expressed by YTC for the development of new process technology to treat fine feed and for the development of new tin products. Arsenic waste treatment was also discussed during the meeting. Certainly more detailed process mineralogy would help improve the YTC operations

Operations at the Mengzi plant were toured and hosted by Li Yu, Chairman of the Board. A polymetallic ore is mined containing lead, zinc, tin, copper, indium, etc. Engineers and staff showed us the plant operations for zinc and indium production. Indium production is one of the highest in China. Products such as lead and tin concentrates are shipped to other plants for processing. The zinc concentrate (600ppm indium) is dead roasted at 850 degrees centigrade in fluid bed roaster, the calcine being leached with sulfuric acid. The leach solution is purified before electrolysis to recover zinc/indium in the cathode. The cathode is melted and then fumed to recover indium in the dust. The dust is leached in acid sulfate and separated by SX using TBP. After stripping with hydrochloric acid the indium is recovered by cementation on zinc plates, cast into anodes and refined electrolytically. After the plant tour, dedication to science and technology was discussed and technology achievements in cooperation with Central South University. Presentations were made by Mengzi and Xuming Wang of the University of Utah. The company was particularly interested in new product development for the indium market in addition to advanced processing technology and environmental issues.

RECOMMENDATIONS

The cooperation between mineral/metal technology sectors in Yunnan is to be commended. It is satisfying to see the close interaction between the Yunnan Department of Science and Technology, the Kunming University of Science and Technology, and the Mineral/Metal Industry (CYMCO, YCT, CCT, Mengzi, Yuntianhua, Feilong Metals, etc.).

Generally, the research and development laboratories need to be improved in all sectors with the exception of the laboratory at Yuntianhua which is excellent. This need is true for the processing of mineral resources as well as for the development of engineered

materials, final value-added products. Considerable research is being done in technology development but greater progress can be made with improved facilities. Not only improved facilities but also continuing education will be needed to maintain and use new facilities.
While there is a clear interest, effort, and progress in technology development, little attention seems to be directed to the fundamental foundation of the technology for processing and product development in the minerals/metals arena. Of course, this is the science portion of "Science and Technology". The science feeds the technology and they must move forward together. As a general recommendation attention must be given and resources committed to fundamental considerations and the application of scientific developments for further progress to be made in the minerals/metals technology.

More specific recommendations follow, including issues with respect to education and research/development.

New/Modified Process Strategies. Efforts to develop new process strategies should be continued to have more efficient use of resources with respect to production, energy, and the environment. Examples include flotation techniques in areas such as phosphate, potash, bauxite, titanium, tin, and zinc especially in the area of fine particle flotation. This includes not only reagent schedule and circuit design but also equipment design. In addition to particle separation processes, hydrometallurgy options should be considered. For example in the case of copper and/or zinc production additional options such as heap leaching and adsorption from slurry need to be examined.

Characterization of Particles and Engineered Materials. Advanced laboratory and plant facilities need to be provided in order to continue to make progress in the minerals/metals technology. One example of such advanced capabilities is the automated instruments for mineralogical analysis, such as QEM/SCAN, MLA, and the new Micro CT for 3D characterization of multiphase systems.

Simulation Tools for Process/Product Improvement. More effective use of simulation tools for process/product improvement should be made. This includes for example both plant and equipment design as well as simulation of fundamental issues associated with the technology, the science of the minerals/metals technology.

Surface Chemistry of Particulate Systems. The foundation of flotation separations is control of the surface chemistry of particulate systems. In this regard, research should be encouraged in such an area using advanced instrumentation to provide a foundation for the technology development. One more specific example is in the area of insoluble collectors and their interaction at mineral surfaces.

Education Matters. Yunnan has a rich tradition of minerals/metals education at Kunming University of Science and Technology (KUST) and such education is critical for future development. The faculty is young and now is a great time for course revision and update. Improvements may be possible by continued training abroad and the participation in short courses. For example faculty/students from KUST might participate in the MODSIM Short Course at the University of Utah, 24-26 June 2009. Other short courses are offered from time-to-time. Also, the University

of Utah would be pleased to have student and/or faculty visit in the spirit collaboration.

In summary, great potential exists for the development of the minerals/metals technology for the Yunnan province. The opportunities are many. Good decisions must be made with dedicated scientists/engineers to take advantage of these opportunities.

ATTACHMENTS

“Mineral Processing and Hydrometallurgy Research"

"Developments in Copper Ore Processing"

Plant tree fossil (on the first page)

Meeting with the heaven and earth's Aura, the essence of the sun and moon, the tree was buried for hundreds million years to become tree fossil and tree fossil jade miraculously. The tree fossil jade has more special values than normal jade does. The tree fossil jade can play an important role in the elegant place. And it is so beautiful that it has highly commercial and collecting value. The beautiful tree becomes tree fossil jade, turning the molder to the miracle, which is the symbol of permanence and success.

Address: 1st floor of bowling hall, Jing Cheng hotel, Ruili City, Yunnan Province, P.R.C.
Phone: 0692-4159886 0691-4159198(fax)
Business time: 09:00-24:00

Tree fossils & Tree fossil Jades (on the third page)

Tree fossils' formation: 150 million years ago of the Mesozoic Jurassic period, the ancient trees were buried deeply under the ground because of Crustal movement. The organic components of the tree were replaced gradually by the SiO2 so as to form the tree fossils, which look like tree, and look like rock too. These tree fossils are heavier than normal rock. The hardness of tree fossil is so high that it is really hard to break them.

The tree fossil jades can date back to 250 million years ago of the Paleozoic Permian period, which are much earlier than tree fossils. The conditions to form tree jade are high pressure, low temperature, and anoxic environment. The main components of tree fossil jades are SiO2, quartz, protein, and chalcedony. Because of the variety of trees and the Geological conditions, the tree fossil jades have the ornamental and collectible value.

In China, the tree species which formed tree fossil jade are pine, cypress, gingko, and so on. In Southeast Asia, the tree fossil jade were formed from robur, redwood tree, such as rosewood, Wenge (also called as chicken wing tree), mahogany, teak.

Categories of tree fossil jade (on the fourth page):

A. sorted by mineral and physical characteristic: quartz tree fossil jade, chalcedony tree fossil jade and protein tree fossil jade.
B. sorted by color: white tree fossil jade, gray tree fossil jade, yellow tree fossil jade, brown tree fossil jade, red tree fossil jade, green tree fossil jade and carnelian-like tree fossil jade.
C. special tree fossil jade: flat and twist tree fossil jade, carbonized tree fossil jade, jasper-like tree fossil jade, tree fossil jade with algae on it, crystal tree fossil jade.

D. the ancient biology can form tree fossil and tree fossil jade, which are really rare. The tree fossils and tree fossil jades are precious unregenerative heritage from the nature, so they have skyscraping collection value.

Burma Tree jade, which can also be called fossil jade, is gaining rising popularity worldwide with its diversified patterns, balmy and fine quality, various color, clear and charming texture and flowing verve.

The TREASURE APPRAISAL column of CCTV has made vivid introduction and favorable comment on Burma tree jade, firmly confirm and promote its values of scientific research, appreciation and collection.

■ Origin of Tree jade

●There is a wonderful legend among Burma folks:

On the occasion of Sakyamuni's nirvana, every creature who believes in him was in deep sorrow, they want to follow him to the heaven. In that very moment, thousands of lights of Buddha emerged, great hurricane took place, Sakyamuni went to the holy heaven while those creatures went to the underground because lack of Buddha power. With the touch of Buddha power, the trees which went underground gradually became glittering and crystal-like, they transformed into wonderful jade, which is the tree jade we talked about. With the hometown of Sakyamuni is in the south of Asia, most of the tree jade is found in some countries in the south Asia. The local folk men always put these tree jade honorably in their living rooms or yards in order to expel the evil and protect themselves, they believe those jades can bring good luck to them, bless the whole family, so that the family can be harmonious and prosperous!

●The formation of Tree Jade:

To put it simply, tree jade is the jade into which tree transformed. During the process of earth's lithosphere, trees are buried deeply under the ground after hot burning or charcoaling process. Hot liquid of magma which is full of Sio2 continually intruded into the tree body, the organism of the trees were day by day taken place by the following elements, like: silicon, calcium, iron, chrome, magnesium, copper and so on. Under strict geological

circumstances and proper geological environment, they have formed the jade-like relic of tree fossil after at least 200,000,000 years. In the field of geology, it is divided into the catalogue of silicon wood, the stone-appreciation circle called it tree fossil or fossil jade, that is the tree jade of our common understanding.

■ Value of tree jade

●First is the formation of tree jade, severe circumstances is highly important, first of all, original forest is required, second is the particular geological element change, such as the volcano eruption, flood or the rock fall, with which the trees are buried into the ground. Third is the suitable burning temperature. Fourth is the fine and proper geological protection condition and circumstance, including enough Sio 2 magma, proper temperature, enough pressure, obdurate circumstance. Fifth is the long time span, at least for 200,000,000 years. Sixth is the singularity. There are a few areas that can form common silicon tree in the world, let alone the origin place of real tree jade! With so large number of collectors, there is few resources can be used.

●The second is the difference of the common jade: tree jade is neither tree nor jade. Its nature has turned into jade, at the same time; it remained many characteristics of trees.

◆First, from the perspective of its appearance, its presence of bark, burl, tree joints, and timber fiber has completely preserved the original flavor of trees. This guileless, natural shape can not be shared by the common jade. As the jade can only present us with its round, random and irregular appearance.

◆Secondly, from the perspective of its section plane, it clearly and completely remains the original annual ring, texture and wormhole of the trees. As usual, we can only see the layer cell, cell wall, water canal of the tree's bast organization with the help of microscope. In contrast, jade is the plain ore with its unattractive and single arrangement of structure.

◆Thirdly, from the perspective of the color, tree jade presents us the colorful world, including the colors of white, grey, yellow, brown, red, purple, green, dark, blue and so on. While the jade can only offer us some colors varying from white to green, there are seldom others.

◆Fourth, from the perspective of resources, the value of tree jade is its unexpected change, various forms, beautiful colors; themselves are different from each other. Every single one of them is the absolute piece of wonder

that you can dream of but seldom have the chance to possess it! While we can easily find two similar jade by usual means in our daily life.

◆Fifth, tree jade is the plant fossil given birth by our great mother of nature, it can be used for other living creatures and geology of that ancient years far away from our present years, therefore, they have precious value of archaeology and scientific research. While jade is used for viewing and admiration.

◆Sixth, tree jade is not only can be made into small and cute handicrafts, they can also be used to make large even giant sculpture. Its value is that we can use it naturally without any formation to decorate our house or yards, or just treat it as the symbol of peace and serenity so as to use it as the home-calming treasure. By contrast, jade can only be made into small things with its carving style and techniques appreciated.

●Thirdly, tree jade and gem. The first requirement of becoming gem is to be completely natural; the second is to be beautiful, it can gladden the heart and please the eye; thirdly is the rigidity to certain amount; The fourth is rare. In all, tree jade has possessed all these requirements, it can be called gem undoubtedly!

■ Appreciation of tree jade

●First, Color

◆1、The best is the one with green color. Green tree jade is very sparse, it gained enough water, has great power of penetration, strong and unshakable construction, it represents life and prosperity.

◆2 、The red one is the ideal one. Red tree jade is also very sparse, at the same time, red represents good luck and fortune, it is the lucky one.

◆3、The blue one is the elegant one. Blue is deep, profound, simple but elegant, it can bring to people a sense of peace and freshness.

◆4、The yellow one is the royal one. The color of yellow is the color of emperor; it implies the nature, grace and high taste.

●Secondly, Nature

◆1、Moist like water. The standard noun people used to test gem is “water”, because its brilliant, its moist and mirror-like appearance, bright outlook and feeling can be tree jade the best of the gem.

◆2、Clear like crystal. The other common used word to describe gem is “transparent”, it refers to its transparency, and it has fine and minute texture, just like a good looking crystal.

◆3、Sculptured veins. Tree jade's veins or its textures represent its internal and external construction, it gives the looker a feeling of great changes of the worlds through immeasurable years. The clearer the annual ring is, the texture is better.
◆4、Mellow like moon. The tree jade which is mellow, clear and bright like the moon is of high standard.

●Thirdly ,The appearance of tree jade: grotesque in shape

The tree jade which has obvious tree joints represents the prosperity in one's life, it is very precious; the one with wormhole in it, especially with the worm's powder after worm's biting in the hole, is of supreme value. Some tree jade looks like certain lovely animal, vivid and good-looking, they are very scarce; some tree jade's arrangement is very clear cut, it possess the grand taste, and brings people relaxation and freshness; those with tree boles are compared to be in the full blossom by the local folk men, its price is much higher than common tree jade; some tree jade are accompanied by aerophyte, such as bines, alga and so on. Those common animals does not include snake, for tree jade with snake is very valuable; and there are still others have been flooded by water for so long time, its mirror-like appearance represents its natural and exquisite lines.

■ value of tree jade

◆1、Value of scientific research. Its appearance represents the annual ring, tree twig, texture, wormhole and tree hull, these can all provide the formation of ancient geology of local land, as well as the weather information, plant and flora.
◆2、The value of appreciation. Tree jades can be said as the permanent paintings, soundless poems, voluminous books.
◆3、The value of collection. The formation of tree jade needs billions of years, there is scarce amount in the world nowadays. It is predicted by experts that there is no resources to be explored after three to five years, its price will rise steeply.
◆4、Value of sculpture. Tree jade has hard texture, vagarious pattern, splendid colors, special textures, it expands the artist's material, species and thought.
◆5、Value to peace protection of the house. Tree jade serves as the treasure of the house, for it can bring happiness and peace to the very house it stayed. Besides, it combines the aesthetic value and its practical meaning, highlights its magnificence, riches and honor. In the long history, it is often

used as the way to help its owner to promote, enjoying auspicious, hope and peace!
◆6、Value of health care. Jade can “exterminate heat, expel worries, moisture lungs, clear the throat, nourish the hair, nurture the five internal organs, quite the soul, dredge the blood, and brighten the ears and eyes”. Besides, tree jade can also nurture people with the own characteristic of wood, therefore, the hearth care value of tree jade is marvelous!

■ Category of tree jade

◆1、Green one, with its bright green color which looks like ice, belongs to the exquisite catalogue.
◆2、White one, with its enough moisture and water-like elements, is the essential parts of tree jade.
◆3、Wormed one, refers to the tree jade with worms, is very valuable. Experts from CCTV channel two’s column: TREASURE APPRAISAL once said that, every single worm in the tree jade worth more than 100,000 RMB.
◆4、Dry one: it has high consistency and tenacity, while transparence and moisture is less than enough.
◆5、Fragile one: with low consistency, changeable texture and color, it may easily crack.
◆6、Water-eroding one: is the plain tree jade washed out for many years, belongs to the common catalogue.

■ Maintenance of tree jade

Comparatively simple, avoid acerbity, alkali, collision, high temperature, and constant soft cloth’s clean with natural water is required.
Tree jade is irreproducible resource. It possesses high value of scientific, archaeological, appreciation and collection. It is indeed a precious present of top grade! This pre-ancient treasure is continually become precious with its economic value rising day by day!
Tree jade, with its heavenly soul and solid form, it gathers the spirit of heaven and earth, combination of gods and auspicious treasure of expelling the evil! Tree jade is the soundless poem and immortal painting!

GREETING and INTRODUCTION

Dear Friends!

Welcome!

The Trinity International Church, was originally named “The Holy Trinity Church.” The old church sited at 218 Wucheng Road (now Renmin zhong Lu) was built in 1903 as the mission station of the China Inland Mission. The site consisted of three parts. The first part was a three-story building with the front

facing to the street. The second and third floors were the church offices and a meeting room on the second floor which could hold thirty to forty persons. The first floor was a hall way leading to the church, which could hold three to four hundred worshippers. At the back was a living quarter for the minister. On 1963, the Christian Three-Self Patriotic Movement Committee of Yunnan Province was formed, and made its headquarter at the Holy trinity Church. On December 1984, the Holy Trinity Church was reopened to worshippers. On 1986, it became headquarter of the Two Christian Committees of Yunnan Province. On February 1997, the church building was torn down due to metro city reconstruction. For eight years, the church met in tents and at different theaters. The believers prayed earnestly for the reconstruction of the new church.

Finally, on June 22nd , 2003, a ground breaking service was held for the new church building under a new name—the "Trinity International Church." The seven-story Gothic style building was dedicated on December 12 nd, 2004. The basement and from the ground to the third floor are used by the Church. From the fourth to the seventh floor are used as the headquarter of the Two Christian Committees of Yunnan Province.

From 1984 to 2006, more than 10,000 people were baptized into the Trinity International Church. Today, every Sunday more then 4,000 believers worship together. In average, every year around 500 believers were baptized into the Church. The Church staff is consisted of three ordained ministers, four ministerial workers, and an accouter.

Thanks for visiting! If you want to know more details, please feel free to contact our English Ushers.

May Grace and peace from Lord Jesus Christ be with you!

Notice:

1. There are three Sunday worship services at the Trinity International Church.

First service Sunday 8:00 to 9:30 AM

Second service Sunday 10:00 to 11:30 AM

Third service Sunday 14:00 to 15:30 PM

2. Christian Bookstore is on the third floor.

Opening schedule:

From Monday to Thursday

8:30—11:30 AM; 2:30—4:30 PM

Sunday

8:30—12:00 AM; 2:00—4:00 PM

3. Offerings and Donations：Donations could be dropped into the Offering Box at corner of the first floor or contact Church Office, Tel: 0871-3639180-403 at 8:30—11:30AM; 2:30---5:30PM

4，Contact us:

Name: Trinity International Church

Mail: Renming Zhonglu Road 188, Kunming, Yunnan Province 650032

Email: sygjlbt@gmail.com

Web: www.trinitychurch.org.cn

Tel: 0871-3616390， 3639180-403

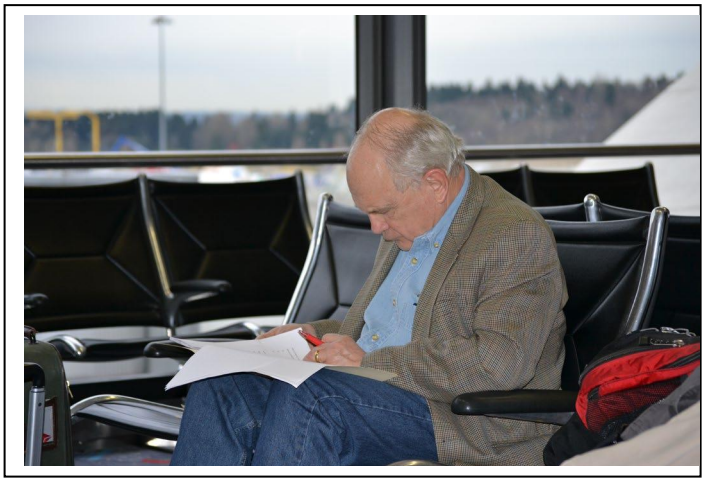
China Journal
4-21 March 2011
J.D. Miller

Preface

This trip to China included stops at the Institute of Process Engineering, Chinese Academy of Sciences, Beijing; International Conference, Beneficiation of Phosphates VI, Kunming, Yunnan; Meeting with Yunnan Department of Science and Technology, Kunming, Yunnan; Meeting with CYMCO, Kunming, Yunnan; KUST (Kunming University of Science and Technology), Kunming, Yunnan; and finally the University of Shanxi, Taiyuan, Shanxi. Of course it was intended to meet many former students and researchers who have participated in research with our group at the University of Utah. In addition, it was planned that new research projects would be initiated during the trip. Previous trips to China were made in 1983, 1997, 2007, and 2009.

Friday/Saturday, 4/5 March 2011

Travel to China with my good friend, Dr. Xuming Wang of our research faculty, began on Friday the 4th with a flight to Seattle. Xuming was not feeling well, having a cold/flu. Also with us was Dr. Jin Liu, a former postdoc in our research group (2007/2008) who had visited with us last month and lectured on spectroscopy to my flotation chemistry class. Jin was the person who had cared for me during the IMPC and my visit to Brisbane, Australia in September 2010. The flight went well and we had a 5 hour layover before departing for Beijing. I had business class accommodations as arranged by CYMCO and was able to sleep a few hours. We arrived at about 9 pm Saturday evening where we were received by Dr. Hao Du, former student (Ph.D., 2007), and Grace, wife of Shili who studied MAC with me for about one year in 2007 as a visiting researcher at the University of Utah. Grace and Shili had visited at Anchor on at least one Sunday during their stay with our research group. Xuming and I stayed at the 5 Star Wenjin Hotel close to the Institute for Process Engineering (IPE), Chinese Academy of Sciences.



JDM correcting papers while waiting for departure at Seattle airport

Grace and Shili Zheng, University of Utah (2007)

Sunday, 6 March 2011

We began the day at the Institute for Process Engineering, Chinese Academy of Sciences, Beijing, being greeted and introduced by Director, Professor Tao Qi, Professor Yi Zhang, Dr. Hao Du, to other researchers. Shili was not able to join us since he had been on assignment in Melbourne Australia. I presented some of our tomography research, "Characterization, Analysis, and Simulation of Multiphase Particulate Systems," to researchers at IPE. The presentation was well received, after which Professor Yi Zhang, member of the Chinese Academy of Engineering, founder of the China National Engineering Laboratory, and well-known scientist in the field of chemical metallurgy in China awarded me with the appointment as Honorary Professor of the Institute of Process Engineering, Chinese Academy of Sciences. After further discussion and lunch we met at Tsinghua University with Professor Yi Jiang and his team regarding energy efficient buildings and the US-China joint research program. This is a program in which USG may have some interest. We then checked out of the Wenjin Hotel. Hao Du took us to the airport for our flight to Kunming, Yunnan.

JDM with Professor Yi Zhang, Institute for Process Engineering, Chinese Academy of Sciences

Sunday was a long day with Hao Du and presentation/meetings at the IPE Chinese Academy of Sciences in Beijing. We arrived at about 10 pm Sunday night in Kunming, Yunnan for the International Conference, Beneficiation of Phosphates VI. They had to upgrade me to a suite in the new 5-Star Empark Grand Hotel, luxurious accommodations for the conference. Xuming was still not feeling well.

Monday, 7 March 2011

The opening session for the International Conference, Beneficiation of Phosphates VI, was Monday morning, with more than 300 delegates in attendance. A spectacular venue at the Empark Hotel had been arranged by our Chinese hosts. Presentations were given with simultaneous translations. Old friends in attendance included Patrick Zhang, conference organizer, of course, John Ralston from Australia, Somasundaran from Columbia University, Michael Baker from Bateman in Israel, Yuehua Hu, Vice President, Central South University in Changsha, and many others. Of course, Yuehua Hu is an old friend, having participated in our research group at the University of Utah (1995 to 1997). The conference was supported by the Yuntianhua Group, the largest phosphate producer in China.

In the morning session, Sandro Marino, my student from Brazil (the soccer player), was recognized and received the outstanding student award for his paper on the use of a magnetic flotation column for the separation of magnetite from a Brazilian phosphate ore. Later, in the afternoon, Sandro presented his paper and did a good job with his presentation. Monday evening we had the conference banquet in the Empark Ballroom. Entertainment was provided by the Yunnan Lantern Theater performers and the Troupe of Arts of Yunnan Phosphate Chemical Group. The entertainment included ethnic dances of the Yi, Tibetan, and Lisu people as well as special contemporary performances. I was fortunate to be seated at the front table and was decorated with special momentos by the dancers. I slept a bit better and expected to be adjusted in a couple more days.

Tuesday, 8 March 2011

Breakfasts at the Empark Hotel were excellent. Today the sessions on geology were not of interest and so Xuming and I traveled to CYMCO's Institute to review research on bauxite ores and zinc concentrates. We met with the new Institute President, Li Xiaoyang, and his project leaders, some old friends who had visited the U of U for research participation. Before lunch at the Jia Yan restaurant we met Wei Xia, the Christian young lady who had been my hostess when I visited Kunming, Yunnan in 2009. After lunch we were returned to the conference at the Empark. Later a meeting was held at 4 pm with directors from the Yuntianhua Phosphate Company to discuss potash flotation. After dinner we went to the Yunnan Arts University for the ethnic dance musical show, "The Impression of Yunnan." We returned to the Empark at about 10 pm.

Post Script: The performance Tuesday night at the Yunnan Arts University, "Dynamic Yunnan," was spectacular. We had a police-escorted motorcade (~5 buses) through the city of Kunming, Yunnan (4 million population) with flashing lights for the half hour trip to the university, both to and from. The ethnic dancing and show was as good as you would see anywhere in the world, very impressive. I was impressed by the "Pilgrimage" and the "Peacock" performances. There are more than 50 ethnic groups that are found in the Yunnan province. It should be recalled that in 2009 I visited some of these ethnic group regions, in particular the Naxis in northwest Yunnan at Li Jiang and the Yulong Snow Mountain glacier, > 5000 meters elevation. It was an exciting, but tiring evening since I was still trying to improve my sleep due to the jet lag.

Wednesday, 9 March 2011

My keynote/plenary lecture was given first thing this morning and was received well with several hundred delegates in attendance. Later in the morning we had a meeting with Yaoji Li, Deputy General Manager of YTH Yunnan Phosphorous Chemical Group. After lunch, I did some work and corresponded with Qiang Yu who arrived in Kunming, Yunnan last night. After dinner Xuming and I met with Qiang to organize our trip next week to the Xishuangbanna area.

Tomorrow we will tour phosphate operations and on Friday we leave to Xishaungbanna. The schedule now looks like,

11-16 March 2011, Xishaungbanna
17 March 2011, Department of Science and Technology
18 March 2011, Kunming University of Science and Technology
18 March 2011, Fly to Taiyuan, Shanxi
19-20 March 2011, Meeting with Fengqin Cheng, University of Shanxi, and Director Pan, Shanxi Water Resources Department
21 March 2011, Flight, Depart Beijing at~11 am.

It looks like an interesting trip to the Xishuangbanna region with stops at the Lan Chang zinc operations and at an iron ore deposit, Hui Min.

Thursday, 10 March 2011

After breakfast at the Empark Hotel, the day was organized with phosphate plant tours and discussion of corresponding operations. Of particular interest was the flotation separation of dolomite ($(Ca,Mg)CO_3$) from the phosphate minerals (apatite, collaphanite). We visited operations owned by the Yunnan Phosphate Chemical Group (YPC) of the Yuntianhua (YTH), a national company, partially owned by the

government, and a major sponsor for the Beneficiaton of Phosphates VI conference being attended. Xuming and I had meetings with Li Yaoji, the Deputy General Manager. Visits today included, Research and Development Center of YPC and the Haikou Beneficiation Plant (2 million tons per year) in the morning, lunch at the Ecological Park, Haikou Mine (wonderful work being done on mine reclamation/restoration), then in the afternoon we toured the Jinning Beneficiation Plant (4.5 million tons per year), followed by a tour of the reclamation area and the Zhendan Geological and Ecological Park at the Kunyang Phosphate Mine. The day, 12 hours later, ended with dinner at the YPC Administrative Building and return to the Empark Grand Hotel. I was exhausted and glad to return "home" to my luxurious suite at the Empark Grand Hotel

We had a good phosphate conference at the Empark Hotel with more than 300 participants. Besides the technical aspects, we had a group photo, jade memorabilia, and Sandro appeared on Chinese, local and national television. The video clip connection was sent to Patricia. This week, I did get to see Wei Xia, the new Christian, who was my hostess during the last visit in 2009 and who took me to the Trinity International Church, with a history of the China Inland Mission at the beginning of the 20th century.

In 2009 we went to the northwest of Yunnan towards Tibet and the Himalayas to visit Lijiang and the Yulong Glacier at > 5000 meters. This visit we will go to the south of Yunnan, to the Xishuangbanna region on the border with Laos and Burma.

Friday, 11 March 2011

The phosphate conference ended today. Xuming departed for Dali by bus to see his family. I will meet Xuming a week from today in Taiyuan, Shanxi, where we will visit at Dr. Fangqin Cheng's institute, the University of Shanxi. Now I will leave Kunming, Yunnan and the attention that has been given to phosphate processing to travel with Qiang Yu. You may recall that Dr. Qiang Yu, former student, Senior Scientist at USG's Research Laboratories in Libertyville, Illinois, and consultant for CYMCO in Yunnan, arranged for this phase of my travel. Qiang met me at the Empark Grand Hotel this morning. After checking out we went to the airport where we met Director Zeng and Guo Hong of the Yunnan Department of Science and Technology (DST) and Vice President Peng of the Kunming University of Science and Technology (KUST). We flew on MU5917 to Jinhong in the Xishaungbanna area of Yunnan. Drivers collected us and we drove south 3 hours on mountainous roads to Lan Chang. This region of Yunnan is the location of the lead/zinc mines and mineral processing plant as well as other operations, such as the iron ore resources. Lan Chang was the location of a devastating earthquake in 1988 with a Reichter number of 7.6. Just yesterday a more modest earthquake of about 5.4 was felt with the epicenter ~ 100km to the west of us. There was significant damage and the loss of ~22 lives.

This area of subtropical Xishaungbanna is the area where tigers use to roam and Asian elephants still abide. The Asian elephants are smaller in size than the African elephants. The Xishaungbanna mountainous country is delightful with banana plantations and terraced tea trees as far as the eye can see. In addition, the valleys have rice paddies attended by water buffalos. I really enjoy the water buffalos as part of the animal kingdom, one of my favorites of the creation. On the drive to Lan Chang we encountered a herd of water buffalo families on the rather modern highway. The families consisted of mother, father, and children about 20 in the herd. The young water buffalos were cute as they followed their parents and their owners to navigate their way on the highway.

On arrival at Lan Chang, we were greeted by President Jian and his staff from the Lan Chang lead/zinc mine, but the highlight was the musical greeting by young ladies of the La Fou ethnic people all dressed in

their classic clothes. They entertained us during dinner this evening and we enjoyed their quite colorful performance. A great deal of preparation had been made with several formal introductions, including President Jian, and a request for my response. From discussion at dinner, I understand that the La Fou people have a Christian heritage. I expect to learn more on this matter tomorrow. We are now in a more remote area of the Yunnan province, about 40 km from the Laos border, without internet connection, and so it may be awhile before I can transmit this portion of the journal. The change in venue was extreme from the 5 star Empark Grand Hotel in Kunming, Yunnan, to rather modest accommodations in Lan Chang, a more remote area of southwestern Yunnan.

Saturday, 12 March 2011

The first stop on the trip to the Xishaungbanna area was Lan Chang, a few 100 km south of Jinhong. The trip was organized by Guo Hong of DST and included Qiang Yu, Vice President Peng from KUST, former CYMCO Research Institute Director, Jian Zang (and daughter Wie Zang, a student studying linguistics in Japan),Yang Zie of DST and others. During this trip we visited the Lan Chang Lead/Zinc Mine Company Ltd, the visit being hosted by President Jiang Shaoping. Tours included the mine site, the mineral processing plant, the zinc plant and the lead smelter. The latter two operations were being serviced for maintenance. Subsequent discussions included mine safety and the need to pump the mine to control the mine water. Selectivity in the flotation circuits was discussed with respect to pyrite depression. The Lan Chang mine has ~2100 employees, 70 % of whom have university degrees. Of these there are 400 senior engineers, and 900 recent graduates. Future plans for the operation include mining a region of the deposit containing copper and the development of a process flowsheet for this pyritic ore containing from 0.8-2.0 % copper. Also of interest in the future is the development of a molybdenum resource that has been found in the Lan Chang deposit. Specific matters for discussion included the following topics.

1) Mineralogical tools and assistance to understand and improve recovery/separation efficiency, especially for silver recovery, and iron contamination of the zinc concentrate. For example the iron contamination of the zinc concentrate appears be due to the presence of marmatite. In this case the iron can not be removed by flotation. The use of XRD, automated SEM systems, and X-ray micro tomography were discussed and recommended for improved characterization of the mineral phases.

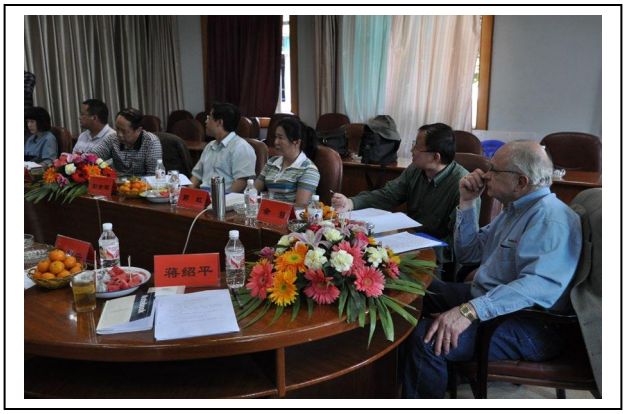
2) Overgrinding is to be avoided. In this regard, a preclassification grinding circuit is being used which is appropriate. In addition, the use of screens rather than the spiral classifiers, or even hydrocyclones, should be considered as a process option.

3) Water quality should be monitored to avoid unintentional activation of sphalerite and/or pyrite by dissolved copper, for example. It seems that this matter is already being considered.

4) The new copper rich pyrite ore contains about 50% pyrite, some zinc and lead sulfides, and 0.5-2.0% copper as chalcopyrite apparently. The grind size for liberation needs to be established and an appropriate strategy for pyrite depression needs to be developed. Possibilities include lime, sulfite, and oxidative treatment. A paper regarding the separation of chalcopyrite from sphalerite will be provided.

5) The future molybdenite ore appears to be suitable for recovery by flotation. Standard flotation techniques should be effective. Attention must be given to the eventual separation from pyrophillite and whether further processing of the molybdenite concentrate should be done to produce a value-added product.

Examination of Lan Chang ore by JDM and Shaopoing Jian, Presdient, Lan Chang Lead/Zinc Mine Company.



Discussion of Lan Chang ore and process strategy.

Today, we heard of the devastation from the large earthquake in the north of Japan and the subsequent tsunamis. All meals were taken at our hotel in Lan Chang which is operated by the local government. I am surviving on a wide variety of Chinese food including frogs, bees, and some unknown dishes. Fortunately there are plenty of peanuts. At dinner we had some further ethnic music of the La Fou people. No internet connection was available at our Lan Chang hotel.

Sunday, 13 March 2011

The morning began with a power failure in the city of Lan Chang and so we had breakfast by candlelight. We then traveled to the village of Hui Min to meet with Kunming Iron and Steel Control Inc. (KISC). Meetings were held with Vice President, Li Ping, Associate Chief Engineer, Luo Ming Feng, and staff personnel of Kunming Iron and Steel Control Inc. (KISC). Revenue for KISC totaled ~45 billion Chinese Yuan last year and this amount is expected to double at the end of the next 5 year plan.

The first meeting was at the site of the new Hui Min ore body, 500 meters deep, 8 km E-W, 5 km N-S, near the village of Hui Min. The Hui Min ore is low grade and varies from 28% to 35% iron as hematite, siderite, and hydrous iron oxides. Quartz is the main gangue mineral but sulfur (~1.2% as pyrite) and phosphorous (~1.2% as apatite) are impurities of concern. A major challenge is the liberation size which is believed to be 10 microns. The Hui Min project is the biggest project that KISC has had in its history. Mineral processing research on the Hui Min ore began in 2006 with both internal and external laboratories participating in the development of a process strategy.

JDM, Qiang Yu, Jinhui Peng (KUST), Guo Hong (DST) and engineers from KISC discuss development of the Hui Min ore body.

JDM and Qiang Yu with Li Ping, VP of KISC, at the Hui Min site.

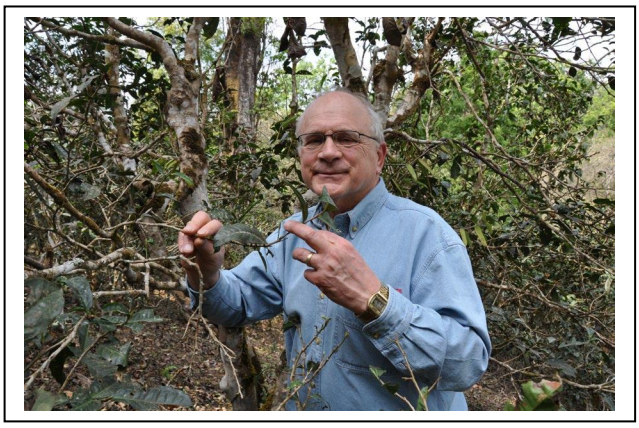
Meetings were continued with lunch at a local restaurant to discuss matters of concern to KISC. The topics discussed were,

1) Production of an enriched iron concentrate. Good results have been achieved in the laboratory but could not be achieved in the pilot plant testing. After some discussion it was decided to review reports and see if the mineralogical analyses were in order and in sufficient detail to guide us regarding liberation. Specifically it was suggested that quartz rejection be accomplished at as coarse a size as possible. The deportment of sulfur and phosphorous would be considered in subsequent testing and in the worst case these impurities would be passed on to three stages in the furnace or to an Ausmelt furnace developed in Australia.
2) Ultrafine grinding. Depending on the desired particle size for liberation, appropriate grinding circuits/equipment should be selected, including a stirred mill as necessary. In order to proceed with this topic the Bond work index for Hui Min ore is needed. It is expected that flocculation of the finely ground product can be achieved for dewatering using some new polymers that are particularly effective for iron oxides.
3) Process alternatives to mineral processing might include thermal treatment, such as roasting of the siderite rich ore to drive off carbon dioxide. Other possibilities might include direct reduction. These thermal treatment possibilities need to have further consideration and such information will be transmitted at a later date,
4) International collaboration. It was suggested that collaboration be initiated by providing copies of reports on process mineralogy, grinding, and flotation of the Hui Min ore for review by research staff at the University of Utah. After review, it is expected that specific action items will be identified in order to move forward in the development of an appropriate process strategy.

Jinhui Peng, VP KUST, Li Ping, VP KISC, JDM and Qiang Yu. Presentation of beautiful tea set for Professor Miller's wife.

At lunch many photos were taken with KISC, with the owner of the restaurant who had never been honored to serve such a distinguished professor from the US, and with a tour guide from the Hani people who was dressed in the ethnic costume. After lunch, the tour guide took us into the mountainous regions at 1500 meters to visit the ancient tea tree forest from 1000 years ago, planted by the Brown people, another ethnic people. As mentioned previously, there must be about 50 different ethnic people. In the Xishaungbanna region there are, for example, La Fou, Tai, Dai, Brown, and Hani. In other regions of Yunnan near Dali and Lijiang (Yulong Glacier), we visited the Yi, Naxi, and other ethnic people in 2009. At the ancient tea tree forest, we collected a special, rare herb, pang hia jiao "crab legs" which only grows in this location.

Special herb, pang hia jiao "crab legs," grows as a parasite in the ancient tea tree forest.



JDM harvesting the pang hia jiao "crab legs" at the ancient tea tree forest.

After the tea forest our guide took us to a mountain village of the Brown people for a tour and discussion of their customs, social organization, etc. The construction of their homes and organization of the village was quite interesting. Lots of chickens; roosters, hens, and their broods.

House construction in the village of the Brown ethnic people.

JDM in the village of the Brown ethnic people.

Woman in the village of the Brown ethnic people.

We then made our way some few hundred kilometers to Jinhong and further east to the Xishaungbanna Tropical Botanical Garden, Chinese Academy of Sciences where we had dinner and accommodations for the night. Our guide from Jinhong was a young lady from the Dai people, Ms.Cheng. The Dai have a strong heritage in Buddhism and rarely travel far from their village.

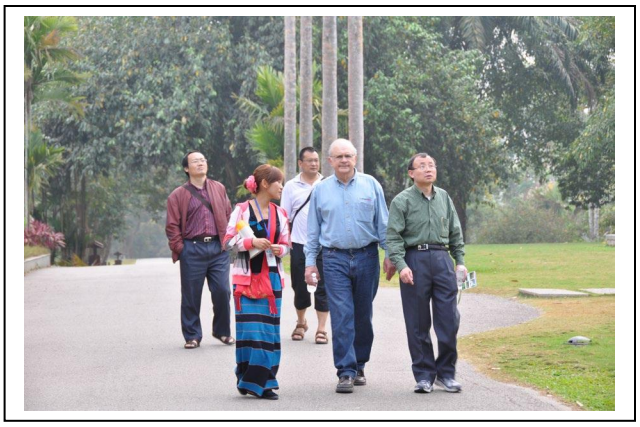
Monday, 14 March 2011

Travel on Sunday had taken us north along the Lan Chang River to Jinhong and then east to the Xishaungbanna Tropical Botanical Garden (XTBG) operated by Chinese Academy of Sciences where we had accommodations at the XTBG hotel.

It is hot and humid today as we tour the Xishaungbanna Tropical Botanical Garden (XTBG). The garden

was founded by Professor Cia Xi Tao in 1959 and is now under the auspices of the Chinese Academy of Sciences. The garden is the largest and most diversified in China with research laboratories and many activities including medicinal studies. Professor Tao died in 1979.

The 1000 hectares garden is a peninsula surrounded by the Luosuo River which with the Lan Chang River drains into the Mekong River eventually flowing through Viet Nam and discharging into the ocean.



Dai tour guide, JDM and Qiang Yu at XTBG.

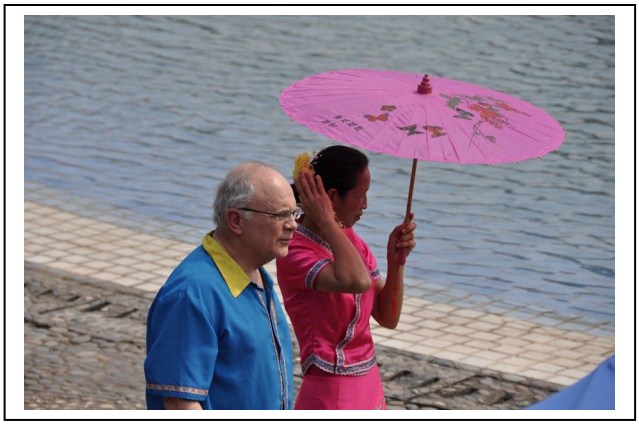
JDM and Qiang Yu at XTBG.

Bee enjoying an exotic flower at XTBG.

Flower at XTBG.

Happy butterfly resting on a flower at XTBG.

After a formal tour of the Garden, Qiang Yu and I rented bicycles and went on our own private tour. Half way around the circumference of the Garden (several kilometers), we came upon a flock of peacocks which was quite exciting. The Xishaungbanna area is well known for its elephants and peacocks. We left the Garden and had lunch at a Dai restaurant. As we traveled we stopped for pineapple along the road and eventually arrived at the Gannabah Spring Water Park where we participated in the water splashing activity. This activity is a little hard to explain but involves some ethnic custom which is really an excuse for a good water fight. Probably ~100 participate in this enormous fountain. The participants include young ladies dressed in their ethnic costumes as well as visitors to the park who dress in costumes provided. Usually it was the boys against the girls. I should be able to get some photographs of this experience. Qiang Yu, Zie Yang, and I participated and we all had a good time.



Ethnic Spring Water Splashing Festival at park in Xishaungbanna Area.

We then returned to Jinhong, checked in at our hotel, the Golden Zone Hotel, and went to dinner at a Tai restaurant with the Mayor of Jinhong. The dinner, which featured Tai roasted chicken, was the best I had since arriving in China.

Tuesday, 15 March 2011

I forgot to mention, last night, Monday night after the Tai dinner, we went to an ethnic show at a theater in the city of Jinhong, population ~ 1M. As usual front row seats had been arranged. I was so tired that I dozed-off through half of the show, but quickly awakened when the Spring Water Festival was enacted with real water. Of course the front row became part of the performance and we all got wet during the water fight. So, on Monday, I really got a complete experience and participation in the ethnic tradition of the water festival on two occasions. Needless to say, I was stiff and sore on Tuesday morning.

After breakfast at the Golden Zone Hotel (we are now back to more comfortable accommodations in Jinhong), we left, traveling by auto, to the Ancient Forest Park about 30 km north of Jinhong. Upon arrival, we attended the animal show, involving tigers, bears, a wolf, a goat, and a monkey. These animals, in captivity, had been trained to do incredible tricks requiring amazing coordination. For example the goat did the tight rope walk, including turn around (180 degrees) with the monkey sitting on his head. Of course, the tigers were huge, 500 lbs., I believe. After the animal show, we toured a bit, had some refreshments, and then watched as 100s of peacocks flew down from high on the mountains surrounding the park lake to land on the shores of the lake. It was feeding time of course. What a beautiful sight as some fanned their tail feathers after feeding.

Wild peacocks at Animal Park close to Jinhong.

Next stop was a Ji Nuo ethnic village. In some regard the Ji Nuo people have a really primitive life style but are trying to adapt to the great changes taking place in China. Unlike the Dai, this people have a patriarchal society. They have no written language of their own. Beauty is ascribed to black teeth and large holes in the ear lobes. Only ~ 20,000 Ji Nuo people are living, not all in this village of course, but the clash of culture is quite evident.

JDM visiting with Ji Nuo ethnic people in the Ji Nuo ethnic village.

Drums used by Ji Nuo villagers in ethnic dance.

We were fortunate on our last stop at Elephant Gorges to see 6 Asian elephants in the wild. Some of our hosts had been to Elephant Gorges 10 times and had not seen the elephants. Qiang Yu is the one who spotted them first. After a tour of the gorges, we saw the domesticated elephants put on a show. Just like being at the circus with front row seats. See photos.

Qiang Yu, JDM, and Jinhui Peng, VP KUST, at Elephant Gorges.

Wild elephants at Elephant Gorges.

JDM feeding cucumbers to trained elephants.

Elephant taking the cucumber bag for disposal.

Another full day with plenty of exercise had been completed and the return to Jinhong and the Golden Zone Hotel was welcomed. We took dinner at a restaurant on the banks of the Lan Chang River. The temperature dropped today and it started to rain.

Wednesday, 16 March 2011

The day began with breakfast at the Golden Zone Hotel, followed by some shopping. Our flight to return to Kunming, Yunnan is at noon. The afternoon will involve check-in at the hotel in Kunming, Yunnan and meetings with the CYMCO Research Institute. I believe a special dinner is arranged with western style steaks from some Kobe beef that had been imported from Japan. To be continued from Kunming, Yunnan.

As expected, I have accumulated so many gifts, mostly of tea, that purchase of another suitcase may be required. In any event we were able to transport everything from Jinhong to Kunming, Yunnan. I should mention that one of the highlights of the elephant show on Tuesday was the soccer match that two bulls put-on. I know everyone would have been impressed, especially Adam and Ryan. I suppose Sandro would have critiqued their technique.

Accommodations are at the luxurious Green Lake Hotel in Kunming, Yunnan, the best in Kunming, Yunnan, and one of the best I have experienced. Quite a change from the accommodations at Lan Chang, Xishaungbana Tropical Botantical Garden, and even the Golden Zone Hotel in Jinhong.

We had steaks for dinner, the beef being raised as part of the Chinese program to have a special breed of cattle to compete with Kobe steaks from Japan. Unfortunately they have a long way to go. The steaks were very tough but we had a good time with friends from DST, CYMCO, and KUST.

Thursday, 17 March 2011

We, Qiang Yu and I, had breakfast in the Executive Lounge of the Green Lake Hotel before departure for our meeting with Long Jian, Director of the Yunnan Department of Science and Technology. Also in attendance were Guo Hong, Yang Zie, and an assistant from DST, Luo Ming Fan of Kunming Iron and Steel Control, Zang Jian of CYMCO, Li of the potash project and Yu Wu, Vice Chair of the Yunnan Chemical Research Institute, both of YTH. I made remarks about the excellent organization that Guo Hong had made for my visit as well as education and research/technology accomplishments for 2009-2011. This presentation was followed by response from those in attendance with corresponding discussion. We concluded our meeting with lunch at a very fine restaurant during which time Long Jian expressed appreciation for my efforts and presented a special scarf for my girl friend. You would be impressed with all the different Chinese dishes that I ate with chop sticks. Qiang Yu left after lunch.

Research Meeting with President Li Xiaoyang, CYMCO Institute

The afternoon and evening were spent with the CYMCO Research Institute in meetings. Ms. Mao served as translator. During the afternoon I presented a lecture on the use X-ray Computed Tomography in mineral processing and hydrometallurgy. About 50 scientists and engineers attended the lectures. Wei Xia came to see me. She is still attending Trinity Church and assured me that she continues with her Christian faith. Dinner was held at the Institute Restaurant with about 10 individuals from the Institute. Qiang Yu joined us for dinner at about 7 pm. After dinner Xiaoyang Li, Director of the Institute, expressed his appreciation and presented a beautiful pewter plate, a momento of the Institute. Earlier he had informed me that we would continue with our collaboration on bauxite flotation research funded by our joint application which had been approved by the central government.

Now I need to pack in preparation for departure tomorrow, Friday, 18 March 2011. Of course with all the gifts it is clear that we will need to purchase a suitcase. I'm not sure when that purchase can be accomplished since the morning is full of meetings, a presentation at Kunming University of Science and Technology, and the Honorary Professor recognition. Well I'm sure we will manage. The Lord has taken such good care of me during this trip with many blessings and I expect that a suitcase will be provided. The flight to Taiyuan, Shanxi is at 2:30 pm tomorrow afternoon.

Friday, 18 March 2011

It has been cold and wet in Kunming, Yunnan. I nearly froze yesterday. The same situation exists this morning. We checked out of the Green Lake Hotel at 8 am after taking some breakfast in the Executive Dining Room. Our driver brought a large suitcase for packing of the treasures that had been accumulated. Qiang Yu left me to depart on his flight back to the US. After arrival at the Kunming University of Science and Technology, the Honorary Professor Ceremony began with President Ron Zhou presiding. Present were students, faculty, and guests numbering ~200. Guests included Li, Guo Hong, and Mao Caizhu from the Department of Science and Technology. Zang Jian attended from CYMCO. After the Honorary Professor had been given, I lectured for about one hour before going to lunch at the President's invitation. Dr. Yixin Hua, head of Metallurgical Engineering, expressed interest in collaboration, as did Professor Dianwen Liu of the Mineral Processing Department. I had already made a commitment to Vice President Peng for collaboration on microwave technology. General agreements for collaboration were signed.

President Ron Zhou (KUST) congratulates Professor Miller on the award of Honorary Professor of Kunming Institute of Science and Technology.

JDM lecture at KUST.

After lunch we left for the airport with Lecturer Wei. The baggage was too heavy and I had to pay 350 Yuan for overweight luggage. I was now on my own as I passed through security. The flight to Taiyuan, Shanxi, was exciting with a rambunctious Chinese baby next to me for the 3 hour flight on the Lucky airline. After collection of the baggage, I searched for Xuming Wang who was to collect me. I soon found him with Dr. Fangqin Cheng and her entourage, including student Jing Jing, from the university and Guo Tianen, Director, Water Resources Department of Shanxi. I was presented with a beautiful bouquet and then driven to my new hotel, the Shanxi Grand Hotel. We had dinner with the delegation that had visited the University of Utah in 2010, including Dr. Junfeng Pan (Bo Pan's father). At the end of the dinner Xuming and I were presented with a friendship album commemorating the visit of the delegation at the University of Utah in 2010.

Greeting of JDM at Taiyuan Airport by Qinbo, Jing Jing and Professor Fangqin Cheng.

JDM with metal sculpture at Shanxi Grand Motel.

Saturday, 19 March 2011

Every day seems to be full and today, in the city of Taiyuan in the province of Shanxi, was no exception. I slept well last night, awakened at 7 am and had to hurry to meet our hosts from the University of Shanxi. Xuming and I met Dr. Fangqin Cheng, her staff, and students, including Qinbo who spent at least one year with us in Utah, for a tour of the campus. Again we were greeted with bouquets of flowers. After a tour of the campus we met with President Yang in a formal ceremony and were given a beautiful commemorative plate of the University. My lecture was then given to faculty and students, about 50 in attendance. This was followed by a tour of the Environmental Engineering and Resources laboratories. We have a collaboration with Shanxi on ground water treatment using reactive barriers under the supervision of Xuming. At lunch we were joined by Fangqin's husband and entertained by students from Fangqin's research group. One group read poetry and the other group sang for us. We enjoyed the entertainment they provided. One song they sang in English was "Yesterday." Dr. Fangqin Cheng continues in her Christian faith and has the Chinese Bible on line. Qinbo tells me that he is considering the claims of Christ and has studied the Bible.

Xuming and JDM at Xhanxi University.

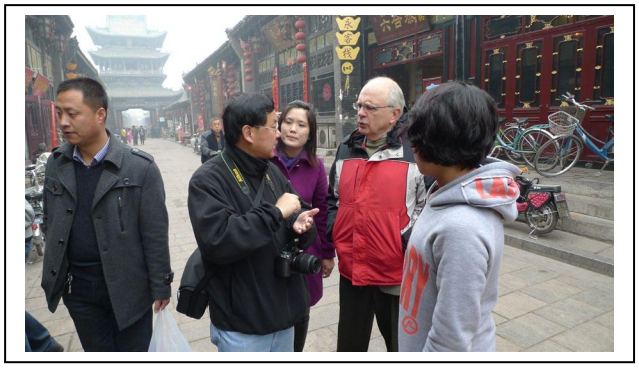
Research group photo at Shanxi University in front of historic entrance.

In the afternoon we met with Director Pan and his staff at the Department of Water Resources. They have developed an impressive system for the management of Shanxi's water resources. After their presentation, we discussed how water quality can be improved for Shanxi. You should know that we will have a three-year project with the Department of Water Resources under the supervision of Dr. Xuming Wang, my colleague, friend, and important research professor in our research group at the University of Utah. We were surprised yesterday when Director Pan had presented us with the special books commemorating the visit of the Shanxi delegation to the University of Utah in 2010. We had dinner at a luxurious hotel with marvelous food and many attendants to greet and serve us. Later after Xuming and I had returned to our Shanxi Grand Hotel, Dr.Fangqin Cheng came to visit bringing more gifts and a request for us to participate in a new research proposal involving soluble salt resources, specifically the processing of potash resources from the Qinhui salt lakes.

Sunday, 20 March 2011

Xuming and I had breakfast at our hotel, The Shanxi Grand Hotel, and then were collected by Guo Tienen of the Water Resources Department and his driver. We drove south from Taiyuan about one hour to the Ancient City of Pingyao, more ancient than Xian. The weather was poor, the air was polluted, and the temperature was about 5 degrees centigrade (41 F). This made the tour uncomfortable even though Xuming shared his coat with me. The ancient city of Pingyao has a history of more than 1500 years. It is famous for having the first banking system known in detail.

JDM having a ride at the ancient city of Pingyao.



Xuming and JDM in a street at the ancient city of Pingyao.

After the tour, we were joined for lunch by Director Pan, his associates, our translator Kelly, the daughter of one of Pan's associates, and county officials responsible for water resources. As usual, we ate at the finest restaurant in the new city of Pingyao. Included on the menu were pig's ears which I had in addition to many other dishes, so many that I can't begin to name them all. I was complimented on my use of chopsticks. After many toasts we finished and soon we were on our way back to Taiyuan. We packed our bags and left for the airport at 4 pm. We were met at the airport by Director Pan, Bo Pan's father (Bo Pan is a student in our research group), Fangquin Cheng and her husband Jiming, as well as Guo Tienen and assistants. More gifts were given and special arrangements were made to carry these with us. Included in the gifts were noodle bowls. I hope they make it safely to SLC. Our flight to Beijing was delayed one hour. We arrived in Beijing at about 9 pm and went to our hotel close by the airport in preparation for our flight home tomorrow.

Monday, 21 March 2011

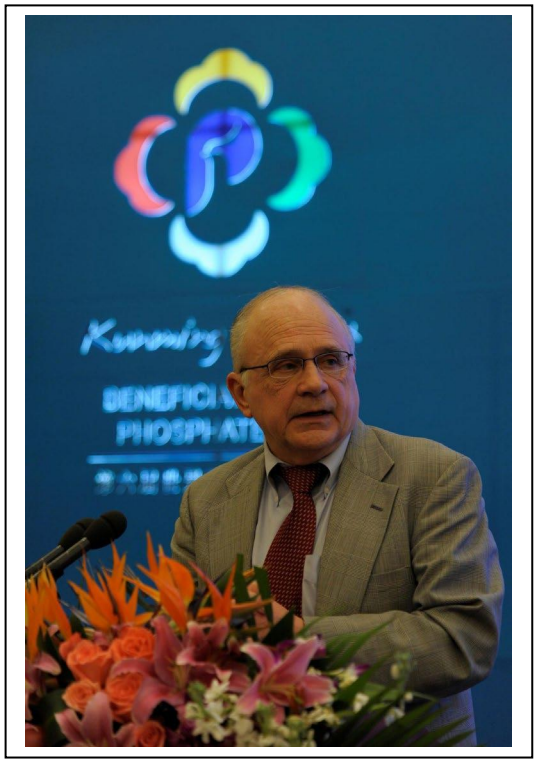
After a night in the Beijing Airport Hotel and breakfast with Xuming, we packed up our luggage and took the shuttle to the Terminal 2 for our DL 128 flight to Seattle. We were met by Hao Du and Shili Zheng from the Institute for Process Engineering, Chinese Academy of Sciences, for coffee before departure. Of course more gifts were provided. Shili is the husband of Grace both of whom attended Anchor Baptist Church several times during their assignment with us at the University of Utah. Grace had recently provided photographs of them and their family. We were able to check our luggage without charge. I think we had twice as much as when we arrived. The trip was quite successful from a number of views. Certainly, as has been the custom, we were treated as if we were royalty and much better than deserved. Our hosts at Beijing, Kunming, Yunnan, Jinhong, Lan Chang, Xishaungbanna, Hui Min, and Taiyuan were so gracious and kind. I am making this entry somewhere over the Pacific Ocean with about 6 hours before arrival in Seattle. We have three hours to clear customs and recheck luggage for our flight to Salt Lake. We thank God for His goodness, protection, and the opportunity to experience more of His creation. It will be great to be home and see if I can find my old girl friend.

JDM and Xuming at Jinning Phosphate Plant YPC, Yunnan, China, 2011.

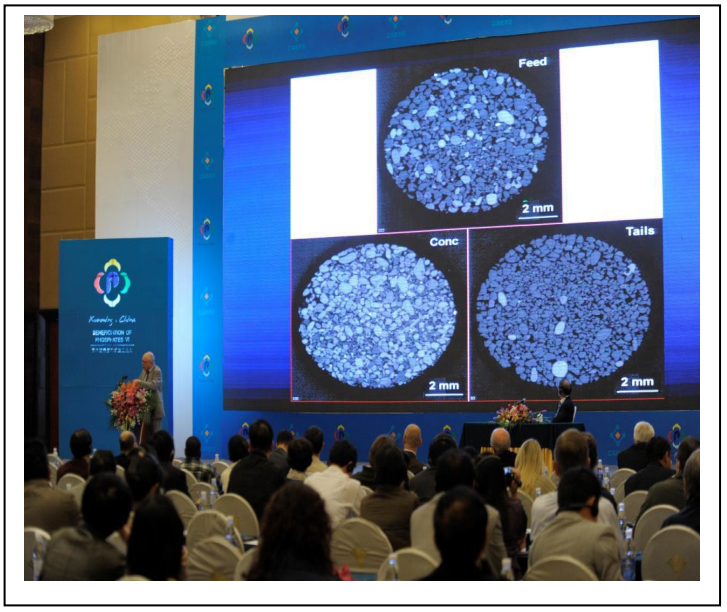
Sandro, JDM, and Sidik with Attendants, Phosphate Conference, Kunming, Yunnan, China, 2011.

Attendants for the Phosphate conference at Kunming, Yunnan, China, 2011.

Participants in the Phosphate Conference, Kunming, Yunnan, China, 2011.



Keynote/Plenary presentation at the Phosphate Conference, Kunming, Yunnan, China, 2011.



Keynote/Plenary presentation at the Phosphate Conference, Kunming, Yunnan, China, 2011.

Journal

J.D. Miller
Mianyang, Sichuan, China
13 – 19 October 2011

Thursday – Saturday, 13 – 15 October 2011

This trip, the second trip to China in 2011, was to participate as the keynote speaker in the 2011 IEEE International Conference on Waste Recycling, Ecology, and Environment at Mianyang, Sichuan. Originally, Dr. Xuming Wang was to accompany me, but Xuming was unable to travel due to his gall bladder operation. After arrival in Seattle from Salt Lake City on Thursday afternoon, 13 October 2011, I waited for 4 hours in the Delta Lounge reading "Walking from East to West" by Ravi Zacharias. Departure for Beijing China was at ~7 pm from Seattle with business class accommodations. We had a good flight and I got some sleep. Praise the Lord. Of course I missed most of Friday, 14 October, arriving at 10 pm in Beijing where I was greeted by Dr. Hao Du, former student and current researcher collaborator. Hao Du took me to the Wenjin Hotel near his institute, Institute for Process Engineering, Chinese Academy of Sciences. I had stayed at this hotel previously in March 2011 when I visited the institute and received an honorary professorial appointment. Hao Du is now married and he and his wife are expecting their first child in December.

My accommodations in Beijing at the Wenjin Hotel were excellent. They upgraded me to a suite, more than 1000 sq. ft. As expected, sleeping was difficult. On Saturday morning, the 15th I read Sadegh's thesis proposal and worked on Sandro's thesis. At noon we (Hao Du and I) went to visit Wang Dianzuo at the Chinese Academy of Engineering. In 2007 Wang Dianzuo took my wife, Patricia, and I to the famous Lao She Tea House for dinner and entertainment including poetry by Li Bai. Of course Wang Dianzuo treated us royally this afternoon.

Later Hao Du took me to the airport for my flight to Mianyang. I was collected at the airport by Dr.Xiao Pengcheng and his student, Jin. We met many conference participants, including Dr. Willis Forsling from Lulea, Sweden, at dinner in the Mianzhou Hotel, where I am staying. Accommodations are not quite as luxurious as at the Wenjin Hotel in Beijing. Meetings for the conference will be at Southwest University of Science and Technology. I speak tomorrow, Sunday morning at 9:45 a.m. as the Keynote Speaker.

Sunday 16 October 2011

The International Conference on Waste Recycling, Ecology, and Environment began this morning at the new campus of the Southwest University of Science and Technology (SWUST), Mianyang, Sichuan, with my Keynote presentation. I suppose that there were 200-300 in attendance. All went well. The day began with thanks to the Lord for his

blessings and breakfast at the Mianzhou Hotel. We traveled by bus to the university. After the morning session we had lunch at the cafeteria. Many old friends from China have come and I have made new friends as well. The day concluded with a banquet at the hotel. I sat with the President of SWUST, Xiao Zhengxue, a classmate of Xuming Wang many years ago at Kunming University of Science and Technology in Yunnan. Mianyang is in Sichuan, the province just to the north of Yunnan. We had excellent entertainment at the banquet with stringed musicians from the SWUST. Professor Fangqin Cheng and two of her researchers from Shanxi University arrived in time for the banquet, As usual she was glad to see me and we had the necessary photographs. I wish the Smith grandchildren had been with me to experience these instruments and the performance. Sleep is still difficult to come by.

Professor Miller, Keynote Speaker, 2011 IEEE International Conference on Waste Recycling, Ecology and Environment, Mianyang, Sichuan.

Stringed musicians from the SWUST at IEEE banquet,
Mianzhou Hotel, Mianyang, Sichuan.

JDM with President Xiao Zhengxue of SWUST and Han Zhilei, a mining engineering student from Beijing General Research Institute of Mining and Metallurgy.

Monday 17 October 2011

After breakfast at the hotel, my host, Xiao Pengcheng and his wife, Lu Xuemei (Wendy), took me to the Earthquake Memorial in Beichuan, about 70 km north of Mianyang, from the basin in Sichuan to the mountainous region surrounding the basin to the north and to the west. Travel took about 1 hour. This earthquake which occurred 12 May 2008 was devastating, reported to have killed as many as 200,000. Other reports state that deaths "only" amounted to 80,000. Whatever the death count, the event was tragic. The Chinese have preserved the devastation as a memorial. Some buildings are crushed and buried with the mountain coming down on them. The first two floors of a hotel were swallowed-up and the third floor was at the new ground level. The courses of rivers were altered, bridges destroyed, other buildings, if not destroyed, were tilted, looking like they were about to fall over. The preparation of the Beichuan Memorial is incredible since these unstable structures have been stabilized with structural steel to provide a reminder to

generations to come. I have no idea if there was any Christian witness in this region. This was a large earthquake, Reichter Scale 8.0, impacting other areas, including Mianyang but particularly Beichuan and Wenchuan counties. They say the devastation in Wenchuan was even worse than what I saw in Beichuan, but fewer people lived in Wenchuan.

JDM at the Earthquake Memorial in Beichuan, site of devastating earthquake on 12 May 2008.

JDM at the Earthquake Memorial in Beichuan, site of devastating earthquake on 12 May 2008.

After return to Mianyang and lunch with Xiao and Lu, I had a meeting at the hotel regarding our micro/nano X-ray computed tomography (CT) facilities with Xiao Wending, friend of Xuming Wang from Chicago days, and with Professor Bao Zhengyu, China University of Geosciences. Bao is interested in creating a Joint Laboratory for Computed Tomography CT and is interested in learning the locations of such systems already in China.

JDM and Professor Bao Zhengyu, China University of Geosciences.

This meeting was followed by another CT meeting with professors from SWUST at the university campus and included Zhang Zhiqi, Environment and Resources, Dong Faqin, Vice President, Peng Tongjiang, Institute of Minerals, Materials and Application. Other professors included, Chen Haiyan, Mengjun Chen, and Liao Qilong. As was the case with China University of Geosciences, these professors from SWUST want to collaborate in the area of computed tomography and prepare a proposal to acquire such a facility. During the meeting Vice President Dong presented me with a book of Chinese stamps with poems by Li Bai, the famous Chinese poet. We, Patricia and I, had gone to the Li Bai Tea House in Beijing with Wang Dianzou in 2007. Li Bai was born in a small town 10 km from Mianyang.

The day concluded with dinner at the Hot Pot, a restaurant where you cook your own food sitting around a pot of boiling oil. Everything and anything goes in the pot, from vegetable to meat including cow's belly. At the end of the evening, it is really exciting to see what you can pull from the pot. I brought back an apron that customers wear to protect their clothes. After dinner we returned to the hotel and I had a good sleep. Praise the Lord.

Forgot to mention that Xiao Pengcheng and his wife gave me a gift of green tea when I visited their home on Monday, 17 October 2011.

Wending and JDM at Hot Pot Restaurant.

Tuesday 18 October 2011

After breakfast at the Mianzhou Hotel in Mainyang, I packed and checked out. Xiao Pengcheng and Xiao Wen Ding took me to the airport at Mianyang for my flight to Beijing at 11:30 am. The flight went well and I was greeted at the airport in Beijing by Director Junfeng Pan, Ms.Chen Bo, and Mr. Tianen Guo, from the Shanxi Department of Water Resources, and at the hotel by Kunlun. They had come to Beijing to meet me, discuss our project on the Cetian Reservoir, provide me with gifts (special box of tea and scroll of cut paper art), and take me to dinner. Of course they had arranged for my accommodations at the Hotel Kunlun, even finer than my earlier accommodations in Beijing. Here I am in the lap of luxury again!! I had a nice swim before a quick meeting and then our dinner of Peking Duck. It may have been the same restaurant that Patricia and I visited in 2007 when we were with Dr. He as guests of Central South University, part of the trip where we visited 6 cities in 12 days. Now it is time to get some sleep in

preparation for a lengthy flight tomorrow with connections in Tokyo and Seattle. Xuming plans to get me at the airport in Salt Lake.

Xiao Pengcheng and JDM at the Mianzhou Hotel lobby prior to departure.

Wednesday 19 October 2011

The last day of the journey was an exceedingly long day travelling from Beijing to Tokyo, to Seattle, and finally Salt Lake City. Director Pan from Shanxi Water Resources Department and his entourage took me to the Beijing airport departing from the Kunlun Hotel at 6:30 am Wednesday Beijing time (4:30 pm Tuesday SLC time). All went well with check-in and with my connections. In fact, on arrival in Seattle, I was able to get an early flight to SLC arriving at about 1:30 pm Wednesday afternoon. I was lucky to find a replacement sandalwood fan for Ginger in the Beijing airport. Ginger had lost her original fan when one of her students broke it. Of course we praise the Lord for his blessings and journey mercies.

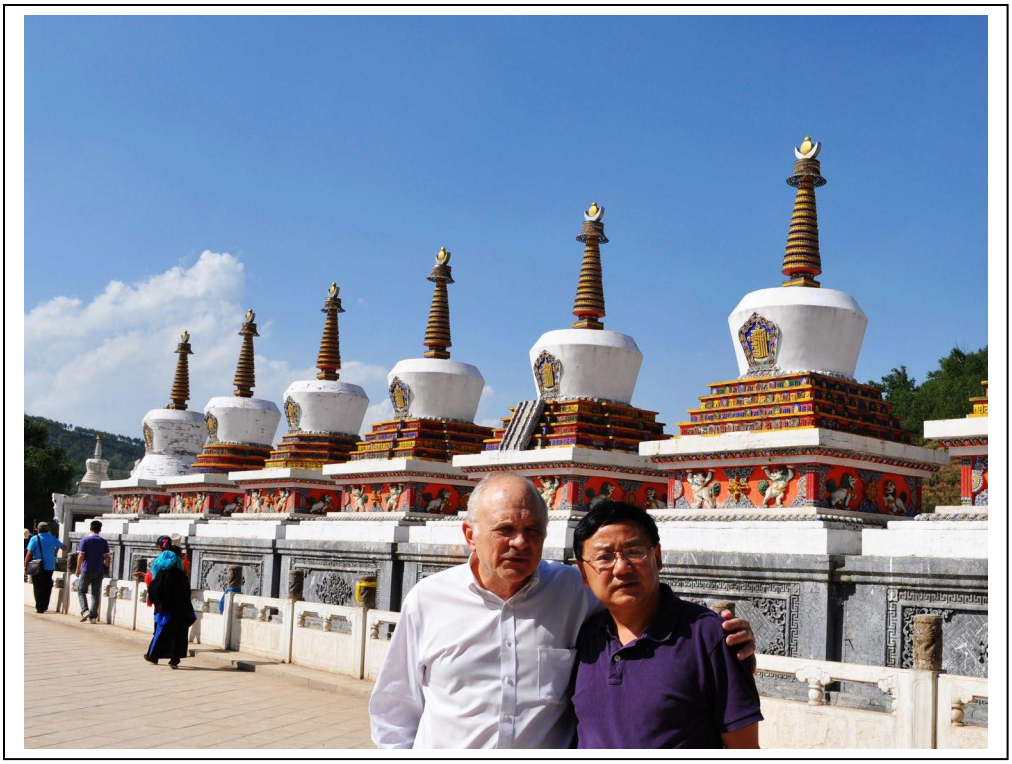
J.D. Miller
Travel to China
Qinghai Province
22 July – 5 August 2012

Preface: Continued travel and collaboration in Western China was accomplished in this trip to Qinghai Province to participate in the conference, "Utilization of Low Grade Potassium Ore," at Qinghai University and an intended trip to Tibet by rail from Geermu to Lhasa. It was planned to visit Qinghai and Tibet because of recent discoveries of mineral resources on the Qinghai – Tibet plateau. These resources range from hydrocarbon resources (oil, gas, and oil shale) to metal resources (copper, iron, lead, and zinc) and will help to satisfy China's needs in the future. See attached story. Also, it was desired to visit Tibet for a number of historical reasons, but specifically it was desired to visit Tibet in search of the elusive, gem quality, red andesine, first discovered in Tibet in 2005 at a small village 300 km southwest of Lhasa. See attached article regarding gem quality red andesine. Unfortunately the trip to Tibet was denied and, consequently, we traveled north about 1000 km from Geermu to Dunhuang, Gansu Province, having opportunity to visit this famous historic location on the "Silk Road." Many of the western provinces have now been visited including Yunnan, Sichuan, Qinghai, and Gansu. Still to be visited are Xinjiang and Tibet, as indicated on the following map of China..

22 – 23 July 2012, Sunday/Monday: Departure at 4:50 pm from SLC on Delta with Xuming, connecting through Seattle, destination, Beijing, China. We had a great connection in Seattle with our gate for the international flight to Beijing adjacent to our arrival gate from SLC. I had business class accommodations and slept well on the flight to Beijing. The flight was not full, so there was plenty of room and the service/food was good. I began reading the *Litigators* by John Grisham, published in 2012. All is well. We arrived in Beijing at ~11 pm on Monday night the 23rd. Of course we lost most of Monday when we crossed the international dateline. Xuming and I were met at the airport by Hao Du and escorted to the Crowne Plaza Hotel. We depart tomorrow morning for Xining in Qinghai province to begin the conferences. The conference at which I speak is, "Efficient Utilization of Low Grade Potassium Ore." After the Qinghai conference, the plan is to visit Tibet then to Taiyuan, Shanxi with a last stop at Beijing before returning to SLC.

24 July 2012, Tuesday: Couldn't sleep very well, probably because I slept on the plane. The accommodations are great. Hao Du, former student, left on the ~7 am flight to Xining. We had breakfast at the Crowne Plaza and arranged for Hao Du's student to collect the Sonar system to be used by Xuming for our research, the survey at Cetian Reservoir in Shanxi, after the trip to Tibet. The Sonar system will have to be shipped or taken to Shanxi for use by Xuming. We didn't want to carry the equipment all the way to Tibet and back.

The shuttle bus took us from the Crowne Plaza to Terminal 2 to catch our flight from Beijing to Xining, about a three hour flight. There at the airport we met Professor Lan Xian from Tsinghua University who was attending the conference. She insisted that I take her 1st class seat. On arrival in Xining we were met by Huaigang Cheng who has been working with us at the University of Utah. He is an assistant professor from Fangqin Cheng's group at Shanxi University, a great guy who has been of considerable help with our research this year at Utah. Also there to greet us were men from the Catia Salt Resource Company (Diao Lin, CEO and Ma Jinyuan, Deputy General Manager). Xuming and I were driven from the airport to our hotel, The Salt Lake Hotel of Xining. After checking in, we left to visit an ancient Buddhist Temple, the Taer Temple, over 600 years old. This temple represents the birth place of Tibetan Buddhism, or Yellow Buddhism. It was here that the 1st Dali Lama was taught. The place is filled with lots of monks, lots of candles, lots of incense, and lots of Buddhas to worship with appropriate rituals, I guess, if you know about these things. I'm not sure how they decide which Buddha to worship. Unfortunately, not too much English is spoken so it was difficult to have a discussion. It seems that this is Tibetan Buddhism which is distinguished from other Buddhisms. I'm not sure what the distinction is, but you probably know that in this form of Buddhism the Dali Lama is revered and I believe that he and the other Dali Lamas are worshipped.



JDM and Xuming at the Taer (Buddhist) Temple, the temple which represents the birth place of Tibetan Buddhism. The 1st Dali Lama was taught here.

JDM examining the Buddhist writings at Taer Temple, Xining, Qinghai.

On return from the Buddhist Temple we met Fangqin Cheng and had a dinner with everyone in the Salt Lake Hotel including the Vice President from Shanxi University, Cao, one of the universities at which Fangqin Cheng is appointed. The other university is the Qinghai University here in Xining. I was exhausted and excused myself early, to retire. Unfortunately the Salt Lake Hotel is not quite what I prefer. It's not the Crowne Plaza. Our rooms are on the 16th floor, the elevators do not run properly, there are no English channels on the TV, and the bed is like sleeping on a board. Perhaps that is the style for the Tibetan Buddhist monks. With this and the unknown Chinese food, it's going to be a struggle to survive the next few days.

25 July 2012, Wednesday: I got some sleep last night and feel a bit better than yesterday. The conference started today at Qinghai University. The campus is new and spacious. They are trying to improve their programs and are making a strong effort to do so with guest speakers, short courses, and workshops. Before Opening Ceremonies we met with the President of Qinghai University, Xi-Dong Liang, and his entourage in a private conference room where he announced my appointment as Honorary Professor in the School of Chemical Engineering, Qinghai University. He then presented gifts to Xuming and me. I reciprocated and gave him one of the magnifying crystals made in Texas.

JDM receiving his appointment as Honorary Professor in the School of Chemical Engineering, Qinghai University.

These activities were followed by Opening Ceremonies for the conference with the President presiding over about 40 in the audience. To my surprise, I was to make my presentation first. The schedule I had been given had me speaking tomorrow. Nevertheless, I survived and the presentation was well received. Of course there were lots of photos.

A Symposium on the Highly Effective Utilization of Low-Grade Potassium Mineral Resources.

Top: Lan Xian, Hao Du, JDM, Ma Jinyuan (Catia Salt Resource Company), Xuming Wang, Fangqin Cheng and other conference attendees.

Bottom: Hao Du, JDM, and Xuming Wang

Later in the day I gave Fangqin Cheng the handcrafted plate from Utah which was received with many thanks. We had both lunch and dinner at Qinghai University. It was a long day and before returning to our hotel, Hao Du took me shopping for some nuts and some new pajamas. It turns out that I left my old pajamas at the Crowne Plaza in Beijing and last night had to sleep au naturale. Tomorrow who knows what will happen? As always I look to the Lord with great anticipation.

26 July 2012, Thursday: Thursday is a rainy day in Xining, a bit unusual for this arid country. President Xi-Dong Liang told me yesterday that this year has been much wetter than is normally the case. After breakfast at the Salt Lake Hotel, full name Salt Lake Hai Run Hotel, we departed in a minibus with other conference attendees, including Hao Du, to the south to visit the Kanbula National Geopark, about a 3 hour drive. Initially the terrain was much like Southern Utah/Nevada with red rocks and arid looking. This turned to mountainous terrain further south as we moved into the Yellow River Basin. The fields were filled with rapeseed with their beautiful yellow blossoms. The rapeseed on harvest is used for oil. This region of Qinghai has a predominance of Chinese Muslims. Clearly we were in rural Qinghai with the people living close to the earth. It reminded me a bit of China in 1983 although even in rural China there are many motorized vehicles today. The Muslims seemed to prefer motor bikes as their mode of transportation.

The Yellow River is one of two great rivers that flow from west to east across China. The head waters for the Yellow River appear to be in Qinghai at a level of 4000 meters above sea level. As the river drops across China numerous dams and hydro power stations have been built to control flooding and to take advantage of the potential energy at that particular point in its course. It was said that 10-20 such dams have been built on the Yellow River. I guess I will have to check this number for a more accurate count.

As we came to Kanbula Geopark, we saw one of the dams mentioned previously at an elevation of 3037 meters, quite an amazing structure. A panoramic view was gained as we drove switch-back after switch-back with our ascent in the park. I'm sure we must have reached 4000 meters. The soft sandstone mountains gave some beautiful views not unlike some of the structures in Southern Utah, but also, because the sandstone was not well consolidated we encountered slope stability problems with slides covering portions of the road.

Top left: Kanbula Geopark dam – elevation 3037 meters.
Top right: JDM, Kanbula Geopark.
Bottom left (L to R): Lan Xian, Tsinghua University, JDM, Hao Du, former student, Institute of Process Engineering, Chinese Academy of Sciences, Kanbula Geopark.
Bottom right: Kanbula Scenic Area featuring Danxia geomorphy.

The Kanbula Scenic Area is known throughout the world for its Danxia geomorphy. It features huge peaks, mesas, caves, and steep cliffs. The mountains take the shape of huge pillars, pagodas, castles, giants and beasts, all diverse and lifelike.

Huiping Song, Huaigang Cheng's wife, took good care of me. We stopped for a roadside lunch and ate from the campfire of the local people. Different foods were available and I had a roasted potato and a peach. During this visit to Kanbula, in addition to enjoying some marvelous scenery, we were able to collect some stones from the Yellow River which are interesting momentos from this visit into the high country of Qinghai province.

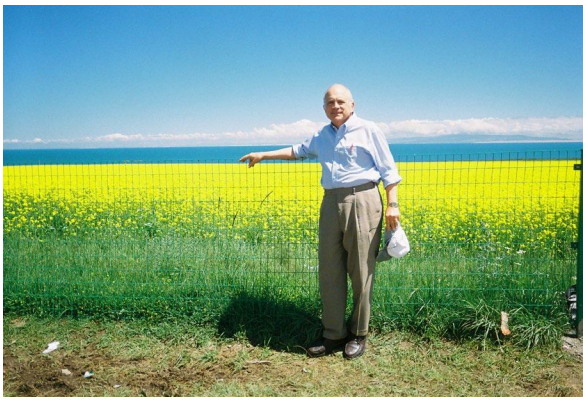
After stopping for a country Chinese dinner, it was the same as you would have in the city, we returned to the Salt Lake Hotel exhausted. Tomorrow we plan to leave Xining and move further west to visit the salt lakes.

27 July 2012, Friday: The big news today is that we have been denied entrance into Tibet and so our plans are changed. I really wanted to visit Tibet, especially by train, crossing in a pressurized train car at 5000 meters elevation. But the Lord knows about these things and I will have to wait for another opportunity. So, our plan is to finish the excursion in Qinghai province, travel north to Dunhuang in the Gansu Province, and then return to Tiayuan via a connection in Xian. Several days will be spent in Tiayuan with Xuming on our Cetian Reservoir project at the Shanxi Department of Water Resources, then on to Beijing to see Hao Du and prepare for the trip home. It's always such a joy to return home, the best part of the trip, any trip.

After breakfast (would you believe fried rice and thoughts of Libo?) at the Salt Lake Hotel, we left Xining (2200m) to travel NW to Qinghai Lake (3000 m), seeing lots of nomadic Tibetan herdsmen, with their tents, minding herds of sheep, goats, and yak. The yaks were the most amazing since we don't see these animals every day. The story is told about a beautiful princess, Wen Cheng, who sacrificed herself to marry a Mongolian King and save her fathers Tang dynasty.

Yak and JDM with goat kid along the highway on the way to Qinghai Lake.

The highway was decorated with the beautiful yellow rapeseed fields for miles and miles, as far as the eye could see. The intense yellow color and the shade of yellow reminded me of our VW super beetle; it would have been well camouflaged in the rapeseed fields. Finally we arrived at the Qinghai Lake which is an enormous and beautiful dead sea but not as salty as the Great Salt Lake in Utah.



JDM in front of yellow rapeseed field at the Qinghai Lake.

Huiping Song, Fangqin Cheng, Xuming, JDM, Cao, Lan Xian, and others at the shore of Qinghai Lake.

From Wikipedia

Qinghai Lake, formerly known as Koko Nur or Kukunor, is a saline and alkaline lake situated on the Tibetan Plateau in the province of Qinghai, and is the largest lake in China. The names Qinghai and Koko Nur both mean "Blue/Teal Sea/Lake" in Chinese and Mongolian. It is located about 100 kilometres (62 mi) west of the provincial capital of Xining at 3,205 m (10,515 feet) above sea level in a depression of the Tibetan Plateau in the cultural area known to ethnic Tibetans as Amdo.[2]

Twenty-three rivers and streams empty into Qinghai Lake, most of them seasonal. Five permanent streams provide 80% of total influx.[3]

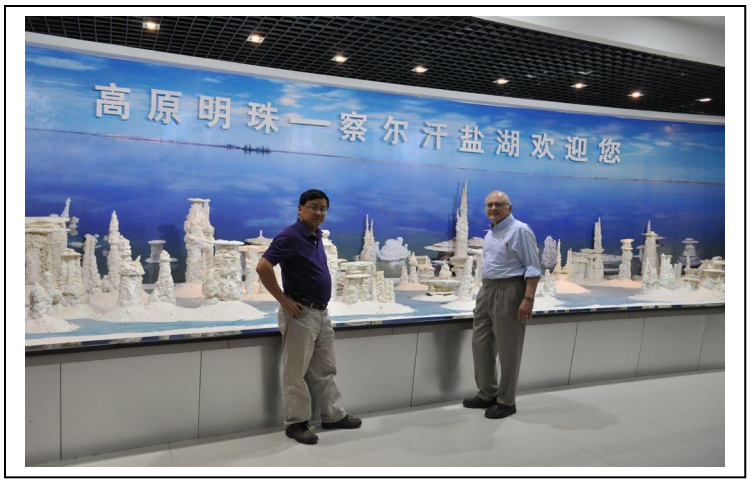
We picked up some stones at Qinghai Lake and travelled on stopping for lunch at a truck stop, again dining close to the earth, nothing fancy but very gracious hosts and all kinds of Chinese food. Our journey today of 800km across the Qinghai province concluded when we reached Geermu (2700m), home of the great salt flats of Qarhan Lake. Geermu is the Chinese spelling, the British named this city Golmud and this is found on some maps. In the same way the British name for the salt deposits in this region is Qarhan, the corresponding Chinese spelling is Chaerchan. Many of the salt and potash operations are in this remote region of Qinghai and tomorrow we will visit the salt museum. Accommodations tonight are excellent considering where we are. They even brought me KFC for dinner! Isn't that incredible?

28 July 2012, Saturday: Xuming and I had breakfast at the Salt Lake Hotel in Geermu and then left with our party including Dr.Cao, Dr.Fangqin Cheng and Dr.Song, all from Shanxi University, and others such as Mr. Ma Jinyuan, General Manager, from Catic, a potash company, and Dr. Lan Xian from Tsinghua University. First we visited the Salt Museum at Qarhan Salt Lake, also spelled Chaerchan Salt Lake by the Chinese. This is the largest and richest salt lake in China and it stretches for 165 km, as I recall. At the plant operations they dredge the solar ponds to obtain a mixed crystal feed material consisting of carnallite and halite. Reverse flotation is then used to separate these mineral components. Carnallite is then converted to potash, KCl, and magnesium chloride. The potash is sold directly, with production to expand to 2 million tons per year. The magnesium chloride is used as feed material to make magnesium metal and selected alloys. Also, chlorine and PVC are produced at the plant site not far, ~1 hour, from Geermu. I'm sure there is considerable history of this unique Qarhan Salt Lake and I will need to research further. But I did learn that it was of early international interest with explorers/researchers from Russia, India, and Sweden making extensive visits in the 1800s.

Top Left: JDM at salt operations for recovery of carnallite from Qarhan Salt Lake in Qinghai.

Top Right: Entrance to Salt Museum at Qarhan Salt Lake.

Bottom: JDM at the shore of Qarhan Salt Lake, in Qinghai, the largest and richest salt lake in China.



Top: Fangqin Cheng and JDM, Qarhan Salt Lake.
Middle: Xuming and JDM at the Salt Museum.
Bottom: Lan Xian, JDM and Hao Du at Kanbula Geopark.

The Kunlun Mountains surround Geermu and the Qarhan Salt Lake to the north and west. If we were to travel to Tibet either by automobile or train we would pass over the Kunlun Mountains through a pass at an elevation of 5010 meters. Maintenance of this special Qinghai – Tibet Railway is costly and difficult. See attached article from the China Daily, Tuesday, 31 July 2012. We did collect stones from the Kunlun River as a momento of the trip. Also some Kunlun jade was secured, courtesy of Ma Jinyuan, to please the ladies in our family. This special jade was used for the 2008 Olympics. I probably need to find some more pieces before I return.

From Internet,

The medals for the 2008 Olympic Games will be made in part with Kunlun jade, which are found in the Kunlun Mountains in Qinghai Province. These will be the first Olympic medals to use non-metal material. But before this, Kunlun jade was sold in the name of another kind of famous jade, Hetian jade, by some dealers who bought Kunlun jade in Golmud city in Qinghai province first and sent it to Xinjiang Autonomous Region. How did Kunlun jade stand out from its competitors and become the material for making Olympic medals? Had it been difficult using jade to make medals? Let's hear the story about Kunlun jade from CRI reporter Liu Bing.

After a good lunch at a western style restaurant, we traveled a bit further toward the Kunlun mountain range to examine special trees that had lived to be 1000 years old, had been dead for another 1000 years and now had been reborn to live another 1000 years. These trees are found in sand dunes to the north of Geermu about 1 hour journey by automobile. We had some excitement trying to take a 4-wheel drive Toyota Land Cruiser across the sand dunes.

Ancient Huyang trees reborn after 1000 years dormancy in the sand dunes north of Geermu.

JDM chowing down a Mongolian dinner at a Mongolian village near Geermu.

JDM at the sand dunes of the Huyang Tree Park north of Geermu.

The day concluded with Mongolian culture and dinner in a round circular home, some were permanent and others were tents. I had a good time with Mr. Ma's daughter, Dou Dou which means bean in Chinese, so I called her Dou Dou the jumping bean. She was trying to learn English and we had a good time at the Mongolian Village; finally arriving back at the hotel at 8:30 pm. Tomorrow we leave Geermu to finish our visit in the Qinghai province.

29 July 2012, Sunday: After breakfast at the Salt Lake Hotel in Geermu, Xuming and I left in one of the Land Cruisers traveling north now with our host Mr. Ma Jinyuan, General Manager of Catic operations at the Ma Hai plant. The weather is cool, windy, and rainy. Mr. Ma, his wife, and daughter, Dou Dou, invited us to lunch at their farm before going to the Mahai plant for discussion of process strategy in the flotation of potash. This section of Qinghai is really remote and desolate, almost a wasteland. Nevertheless Ma has been able to establish a business growing the herb "Goji" a health food that Xuming's wife, Minhua Li, takes on a regular basis. We proceeded over a 3335 meter mountain pass before arriving at the farm. These people live close to the earth and everything that was served had been grown on their farm. I struck-up quite a friendship with Dou Dou, the yellow jumping bean, and am now considered to be "Grandpa." Yellow was a new word Dou Dou had learned since she was wearing yellow rain apparel. You see I still have not lost my rapport with the girls.

JDM, Dou Dou, “the yellow jumping bean,” and father, Mr. Ma.

After lunch we journeyed on to Catic’s Mahai plant and had a tour of operations. Of course we are now 400 km from the Qarhan Salt Lake and into a different potash region. The process involves decomposition of carnallite and as a result fine KCl particles must be separated from NaCl particles and clay particles by flotation. This is a challenge because of the small particle size. We had a good discussion, thanked Ma for his hospitality and left to finish the rest of our journey, but not before I got a big hug from Dou Dou.

We continued north with the land becoming even more desolate. Sand dunes are everywhere. This looks like an extension of what I imagine the Gobi desert is a bit further north in the Xinjiang Province and in Mongolia. I think we saw a handful of vehicles during the next 300 km. Our destination was Dunhuang in the province of Gansu. So as we scaled another pass (3648 meters) in this high desert country, we left the province of Qinghai and entered Gansu province.

At the pass summit the temperature had dropped to 6 degrees Celsius. By the time we reached Dunhuang the temperature had risen to 30 degrees Celsius. Dunhuang is a bit of a tourist town/city and we will be at the Sun Village Hotel in Dunhuang for 2 days. Great accommodations here at the Sun Village Hotel, not quite so close to the earth as in recent experiences during our travel in Qinghai. Dunhuang is known as an important stopping point, an oasis for caravans on the Silk Road trade route which continues on east to the ancient capital of Xian.

From Wikipedia

Dunhuang (Chinese: 敦煌; pinyin: *Dūnhuáng*, also as simplified Chinese: 炖煌; traditional Chinese: 燉煌; pinyin: *Dūnhuáng* in ancient times meaning 'Blazing Beacon') is a city (pop. 187,578 (2000)) in northwestern Gansu province, Western China. It was a major stop on the ancient Silk Road. It was also known at times as Shāzhōu (沙州), or 'City of Sands',[1] "or Dukhan as the Turkis call it."[2] It is best known for the nearby Dunhuang Caves.

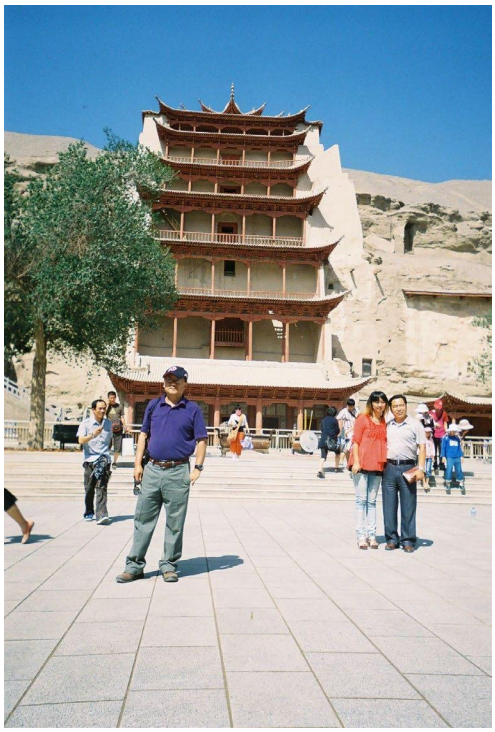
It is situated in a rich oasis containing Crescent Lake and Mingsha Shan (鸣沙山, literally "Echoing-Sand Mountain"). Mingsha Shan is so named for the sound of the wind whipping off the dunes, the singing sand phenomenon.

It commands a strategic position at the crossroads of the ancient Southern Silk Route and the main road leading from India via Lhasa to Mongolia and Southern Siberia,[1] as well as controlling the entrance to the narrow Hexi Corridor which led straight to the heart of the north Chinese plains and the ancient capitals of Chang'an (today known as Xi'an) and Luoyang.

We had dinner at the Market place in the city center and enjoyed the vendors and their wares, lots of produce from surrounding farmers. Then we returned to the Sun Village Hotel and I retired and gave thanks to the Lord after a long and interesting journey. Now, at the end of the day, I'm exhausted.

JDM at the Market Place in Dunhuang.

30 July 2012, Monday: Today was filled with Buddhas, swimming, and camels; and I thought we would have some rest! The Lord gave me a good sleep last night for which I am thankful. In the morning Xuming, Gong, and I visited the UNESCO heritage park, the Mo Gao Grottoes, over 600 caves prepared in sandstone and dedicated to honoring the various Buddhas, some dating back to the time of Christ. At this park the Buddhas are not worshipped, at least we saw no evidence of that. After visiting a few of the grottoes it got a bit old.



Xuming Wang at the Mo Gao Grottoes, a UNESCO heritage park.

Lunch was at the Grand Soluxe Hotel in Dunhuang where they took an hour to prepare this hamburger. It tasted great, along with high quality French fries; who would have thought such would be possible in Dunhuang at their fancy upscale hotel!

We returned to our Sun Village Hotel and had a swim in an enormous pool. It looked the size of a football field. Granny will think that I embellish. Oh, alright, I exaggerate, but it was at least twice the size of the natatorium at the University of Utah.

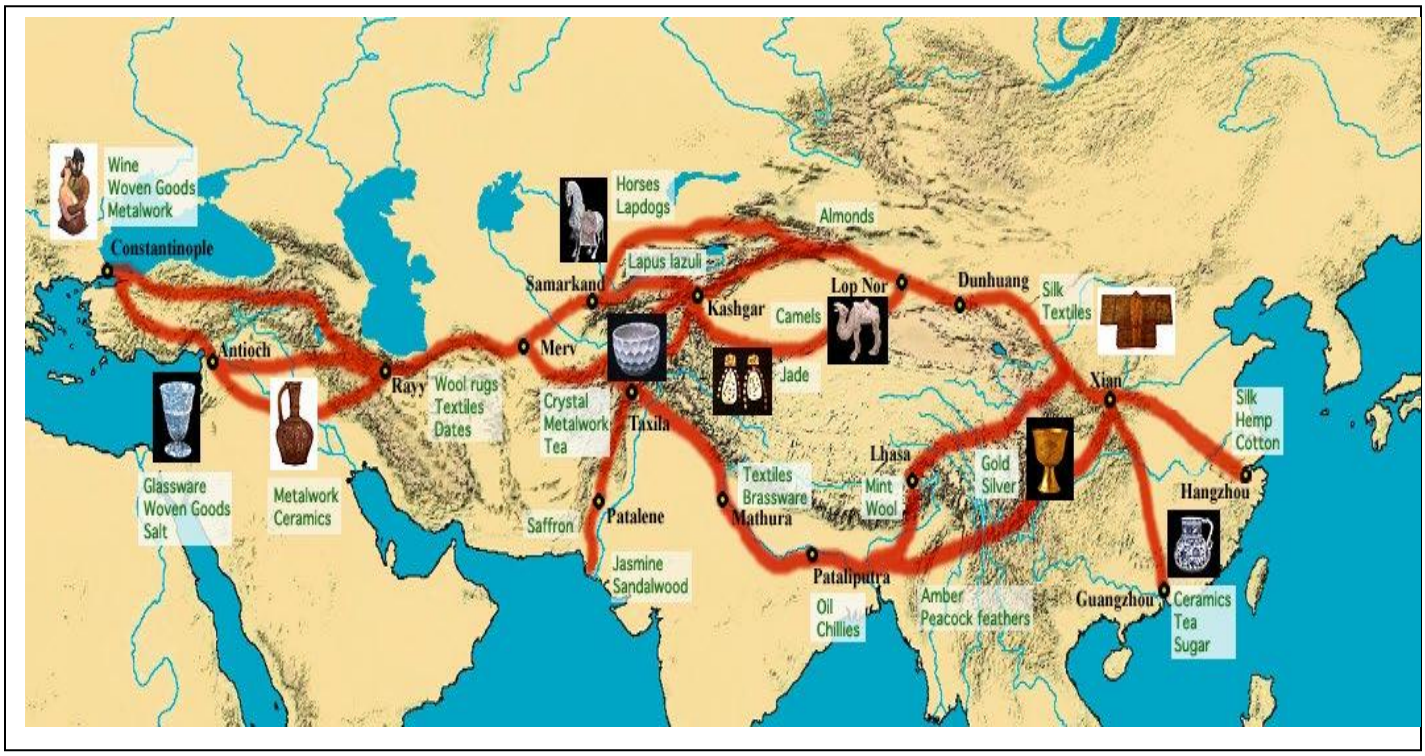
I thought we were done for the day but was informed that we must go to the Crescent Moon Spring. Little did I know that this excursion involved a camel caravan through the sand dunes and the eventual discovery of this special oasis used by traders centuries ago on the Silk Road. It appears that at Dunhuang the caravans could take a southern route or a northern route. The former must pass through or around the Himalayas, through Persia, on to the Mediterranean and north then to Istanbul. The northern route had to contend with the Gobi desert, I believe, and then pass on to Istanbul going across just to the north of the Caspian Sea and the Black Sea. Anyway, the sand dunes at Dunhuang would

knock you out if you could survive the camel ride. This was a problem for me since I have not much padding on my underside. I hope the camel felt better than I did after our journey. The sand dunes were just incredible. Their size still amazes me.

JDM amongst the camels at the sand dunes of Dunhuang. From this location a camel caravan was taken across the sand dunes to Crescent Moon Spring.

JDM and his camel caravan buddy heading across the sand dunes to Crescent Moon Spring.

JDM at the Crescent Moon Spring oasis.



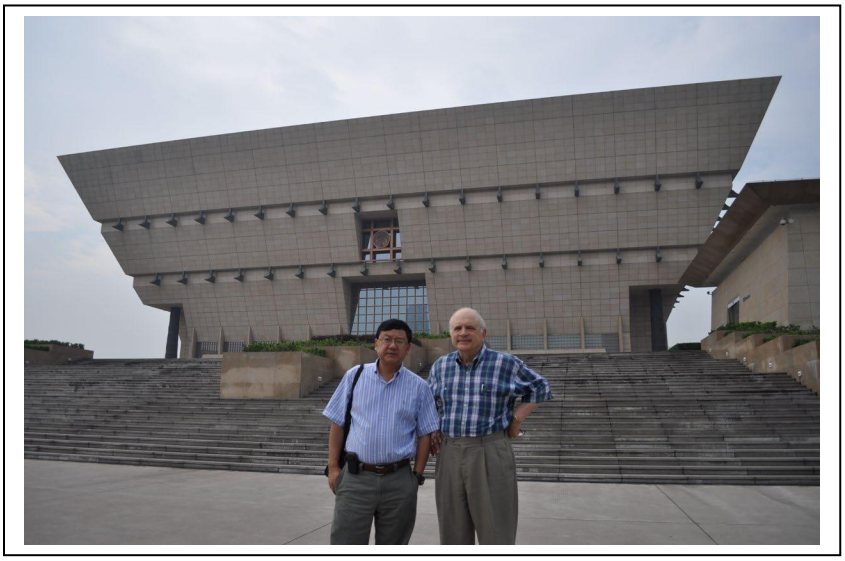
Routes of the Silk Road as they branch out from Dunhuang, China.

Anyway, I finally made it back to my room. Nothing scheduled for tomorrow except a flight to Taiyuan in Shanxi province where we will spend a couple days with my faithful companion and colleague, Dr.Xuming Wang, working on his research program with the Department of Water Resources in Shanxi.

31 July 2012, Tuesday: Today we left Dunhuang and are making our way back to the east with a China Eastern flight to Xian, connecting with a South China flight to Taiyuan. We want to thank the employees of Catic, Ms.Gong and Mr.Wu, who served as our hosts as we travelled from Geermu to Dunhuang. Before leaving Dunhuang I was able to secure silk scarves with a map of the Silk Road, a nice momento of the trip to Dunhuang and western China in general. Not too exciting today but we did meet a young girl who engaged us in English. The trip took most of the day. I discussed research with Xuming and read some more of John Grisham's book, *The Litigators*. We were collected at the Taiyuan airport by Mr.Guo of the Water Resources Department for Shanxi and taken to our hotel. We will be in Taiyuan for several days before flying to Beijing where at the Crowne Plaza I will prepare for departure to Salt Lake City.

1 August 2012, Wednesday: Not too exciting today. Xuming and I have magnificent accommodations in Taiyuan at the Tianrui Business Hotel as arranged by Director Pan of the Shanxi Department of Water Resources (DWR). Essentially suites were arranged for us and we are very comfortable. This morning I met with Bo Pan to discuss corrections for his thesis. For lunch, Director Pan and his wife invited us to go for lunch at quite a luxurious private restaurant that had a wonderful décor. I think I had the best steak I ever had in China, except maybe for that Kobe steak we had for a dinner one night in 2009 when we visited Lijiang in Yunnan Province with Qiang Yu to see the glacier and the Naxi ethnic people. In the afternoon we met with the Water Resources people to review Xuming's project and plan the research program for the rest of this year. The meeting went quite well, but Xuming has a lot of responsibility. In the evening the DWR people joined us at our hotel for dinner which included well prepared pig's feet, and now I'm off to bed in a few hours.

2 August 2012, Thursday: After breakfast Xuming and I were taken by Bo Pan, a current University of Utah student who is finishing his thesis, to visit the Shanxi Museum in Taiyuan. We spent the morning at the museum with a private tour guide, a young graduate of Shanxi University named Marissa, having satisfactory English. We had some opportunity to discuss evolution and Buddhism with her. She was not a Buddhist and was uncertain how many gods they have. Lunch was with Bo Pan and Mr.Gao of DWR at something like a private club. The venue was a bit like Benni Hana where the cooking on a grill is done right at your table. Discussion centered on Bo's girlfriends and when he was going to marry. His Mother is keen to see this happen and it seems that marriage might be possible next year.



Xuming and JDM, Shanxi Museum in Taiyuan.

We then drove to the Lihua Grand Hotel, quite a luxurious hotel with the Presidential Suite available at a rate of $10,000 per night. We settled for a room somewhat less expensive. After a bit of a rest we went for a swim in the Lihua Grand Hotel's enormous pool. Since we did not have our swimming gear, new gear had to be purchased. Included in this purchase was the purchase of prescription goggles so that I could see where I was going. We had a great swim. After a shower, rest, and viewing the Olympics, we went for dinner at the Lihua Grand Hotel before returning to our base hotel for sleep and preparation for departure to Beijing tomorrow. Again, the Lord has provided a most interesting day. Unfortunately I didn't get any of my university work done today, but have high hopes for tomorrow morning.

3 August 2012, Friday: After breakfast, I did some of my university work at the hotel in Taiyuan. We went for lunch at KFC with Xuming, Bo Pan, and DWR people including Mr. Gao, Bo Pan's mother (Gao Xia Men) and others (Lu, Chen, Hu,…..). Gao Xia Men presented me with a beautiful book about Confuscious. In the afternoon, Mr.Gao took me to the airport to catch my flight to Bejing. Xuming remained in Taiyuan. All went well and I was met in Beijing by Hao Du's student, Ying, and her boy friend, Xinyuan. They took me to the Wenjin hotel near the 4th circle and the Institute for Process Engineering, Chinese Academy of Sciences as well as being close to Tsinghua University. We had dinner at the hotel and I enjoyed a nice sleep.

4 August 2012, Saturday: I had a great sleep last night. This morning Hao Du came to take me to his institute (IPE) after breakfast and after checking out of the Wenjin Hotel. This region of Beijing is known as Vzong Guan Cun, a technological region having about

10 universities and many research institutes. Before leaving my room, I caught a CNN documentary on alleged child abuse at Fairhaven Baptist Church which had many interviews, including Pastor Voegtlin and former students. Strong accusations were made. Evidently, there have been demonstrations by citizens of the community for Fairhaven to stop their practices of corporal punishment, humiliation, and more serious claims. About six former students were interviewed. The pastor's adopted son and daughter were interviewed. The son had not spoken to the pastor for ~25 years. He claimed that in addition to being beaten he had been made to wear a dress to school because he was a sissy and had not lived up to his father's expectations. It certainly was a damaging and discouraging documentary. Evidently this was a report on activities that happened in 2011.

At IPE we worked on some of the molecular dynamics simulations research and corresponded with Xia Zhang, my student back in Utah. Later we went for lunch at Quanjude Roast Duck. Of course our roast duck was certified and had its own number (247880). The lunch was delicious and we were hosted by rare earth researchers from Tsinghua University, Linyan Li and Feng Xiujuan. Tsinghua University is maybe the most prestigious university in China. These ladies are associates of Professor Zhenge Xu of the University of Alberta who is on a three year assignment at Tsinghua University but could not join us since he was out of Beijing today.

After lunch Hao Du took me for check-in at the Crowne Plaza, close to the airport. The day concluded with some work at the Crowne Plaza, a swim, and preparation for an early morning departure.

5 August 2012, Sunday: Not much sleep last night. The morning started early with a 5:30 a.m. wakeup call. I had breakfast at the Crowne Plaza and then departed with Hao Du for Terminal 2 at the Beijing Airport. All went well with check-in. I finished John Grisham's *The Litigators*. I really enjoy Grisham. His new books this year are *Calico Joe* and *Racketeer*. The flight is ~11 hours and, in addition to finishing *The Litigators*, I watched 2 movies and tried to sleep without much success. We arrived early in Seattle and I caught an early flight to SLC, 1st class upgrade. Praise the Lord! Patricia assured me, when I called her from Seattle, that she would collect me at the airport. I don't imagine that my luggage got on the early flight so I will need to wait for it or return later to the SLC airport. In fact, Delta agreed to deliver my suitcase to our house and it was received late Sunday afternoon.

It was a great trip, long and somewhat disappointing since I didn't make it to Tibet, but that will give an excuse to return. In any case, the Lord prepared the way and I had a safe and most interesting time.

Desertification threatens Qinghai-Tibet Railway

Updated: 2012-07-31 08:07

By Xinhua in Xining (China Daily)

China's Qinghai-Tibet Railway, the world's highest rail system, is being threatened by desertification on the Qinghai-Tibet Plateau as a result of global warming, experts concluded after conducting a probe.

About 443 km of the 1,956-km railway are in areas affected by desertification, including 103 km that lie in seriously desertified areas, Wang Jinchang, a senior engineer with the Qinghai-Tibet Railway Co, said on Monday.

Wang cited research showing that the threat of soil erosion grew very fast in recent years, mostly near rivers and wetland from Golmud and Lhasa, and the amount of affected track almost doubled from 2003 to 2009.

Touted the "Road to Heaven", half of the Qinghai-Tibet Railway was built on areas at an elevation of about 4,000 meters, crossing mountains, ravines, the Gobi Desert and frozen earth, among other hostile environments.

An Fengjie, an official with China's State Forestry Administration who specializes in soil erosion control, said the plateau region suffered from desertification long before the railway was built.

"The railway did not cause the problem, but it gives us an opportunity to witness the severity and scale of soil-erosion on the Qinghai-Tibet Plateau," An said.

Sands buried tracks and disrupted train services more than 1,362 times from 1984 to 2002 on the Xining-Golmud section of the railway, which has been operating since 1984. The main part, the Golmud-Lhasa section, went into operation in 2006.

Since becoming fully operational, the railway has transported 52.76 million passengers, according to the railway company's estimate this month. Work has begun to expand the railway from Lhasa to Xigaze, a historical Tibetan city and home to the Panchen Lamas.

Engineers set up barrier walls or simply placed big rocks along the tracks in affected areas to prevent sands from encroaching on the rails.

"These emergency control measures have been effective, but we still need to address the root problems of desertification," An said.

One of the most prevalent theories blames global warming for ecological deterioration in the plateau region.

Sun Zhizhong, a researcher with the Chinese Academy of Sciences, said temperatures on the Qinghai-Tibet Plateau rose by more than 2 C on average over the past three years, leaving large chunks of frozen earth to defrosting.

The moisture is soon lost, however, as water quickly evaporates under the plateau's blazing sun. The soil begins to dry up and eventually becomes desert, Sun explained.

He said overgrazing is another concern.

"Lab results tell us that desertification is a trend that will be hard to reverse in the near future. The Qinghai-Tibet Plateau may one day become a major source of sandstorms," said Qu Jianjun, another researcher with the Chinese Academy of Sciences.

Qu said glaciers are also melting quickly on the plateau. More than 200 small glaciers in the region have disappeared over the past five decades. Lakes in the heart of the Hoh Xil Nature Reserve, home to 70,000 Tibetan antelopes in remote western Qinghai, have become swollen as a result of melting glaciers and snow.

Xiao Penghu, deputy chief of the nature reserve administration, described melting glaciers as "a hanging dagger".

He said that after most glaciers melt away, the lakes and rivers in the Hoh Xil region will start to dry out, resulting in "disastrous" consequences for the local ecology.

"The fate of the Tibetan antelopes is in our hands, because our actions set the pace of global warming," said Tseten, another official with the nature reserve, calling for people around the world to unite to fight climate change.

"Once global warming wipes out (Tibetan antelopes), we'll eventually be wiped out as well," he said.

(China Daily 07/31/2012 page7)

Natural Red Andesine from Tibet: Real or Rumor?
GIT Gem Testing Laboratory (GIT-GTL)
3 November, 2010

Andesine is one member of the plagioclase feldspar solid solution series which contains between 30-50 % anorthite component. Andesine is rarely found as gem-quality stone, as compared with the more common species such as albite, oligoclase (sunstone) and labradorite, which have been discovered and mined for long time from various sources.

The first discovery of gem quality red andesine was reported from the Democratic Republic of Congo in 2002 (Krzemnicki, 2004). The first article of andesine from Tibet was reported by B.M. Laurs in 2005. Later, large amount of red andesine stone from Tibet was introduced to the gem market by some Chinese dealers under the trade-name of “Tibetan sunstone” or “Tibetnite”, nonetheless the source where this gem material originates from has still been doubtful. In the meantime, the rumor of treated red andesine had spread out through the whole gem market which created a big controversial and confusion among gem traders worldwide. In 2008, the copper-diffusion treatment of this gemstone was proved experimentally (Emmett et al., 2009) and the report of such treatment was published. Later the technique was confirmed by other study (Milisenda et al., 2008). After the disclosure of such treatment technique then the important questions were arisen among the gem dealers whether the red andesine sold in the markets are the natural or treated ones, and the location or deposit where the natural red andesine was found and supplied to the trade is really existing in Tibet.

In order to prove the existence of this gem locality, **the group of international team consisting of representatives from LMHC member institutes, i.e., Ahmadjan Abduriyim (GAAJ), Brendan Laurs (GIA) and Thanong Leelawatanasuk (GIT) together with independent experts, i.e., Richard W. Hughes (a well-known gemologist), Flavie Isatelle (research geologist) and Young Sze Man (Jewellery News Asia, Hong Kong) with the support from Christina Lu (M.P. Gem Corp.), Li Thong and Lou Li Ping (Tibet Andesine Co.)** had made a field trip to the mining area in Tibet during 26th September – 2nd October 2010. The expedition began in Lhasa the capital of Tibet. The red andesine mines are located in Zha Lin and Yu Lin Gu areas near the border of Bainang and Gyanze counties, about 55 road-km southeast of the Xigaze (Shigatse) city and around 300 road-km southwest of Lhasa city (see Figure 1).

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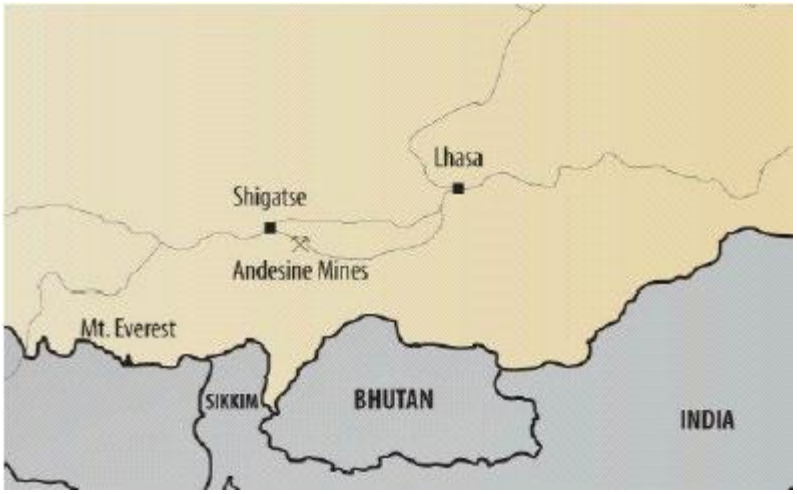

Figure 1: A map showing the location of the andesine mines of Zha Lin and Yu Lin Gu areas (modified from Richard W. Hughes)

According to the headman of the Zha Lin village (age about 50+), who had heard about the presence of red stone in this area since he was just a young boy, however during that time he and other villagers did not know what this kind of gemstone was. Until 2005, after the news of the red andesine mining in Nai Sa, Bainang was spreaded out the whole counties then many villagers began digging and collecting those red stones for commercial purpose. (See Figure 2)

Figure 2: Headman of Zha Lin village told the story of red gemstone to the group (left) and shows his own collection samples (right).

Our team found that andesine in this area occurs in secondary deposits of alluvial material and alluvial fan debris (see Figure 3). We don't know where the andesine within these sediments was derived. It is possible that the andesine-bearing soil at Zha Lin could have originated from the weathering of rocks that have been tectonically displaced.

Figure 3: Overview of Yu Lin Gu valley where the red andesine samples have been found. (top photo: a shot from valley to Zha Lin village, bottom photo: a shot in Zha Lin Village about 1 km from the top photo).

At the Zha Lin village, the sediments consist mainly of silts, sands and gravels of shale and slate. The red andesine samples can be found scattered on ground surface to about 1 m depth in the digging pit (see Figure 4).

Figure 4: The digging pit near Zha Lin village (top), and red andesine collected from the pit (bottom).

In order to be sure that these gemstones do occur naturally, our group has attempted to randomly select the digging sites around the area without assistance from any local villager in order to rule out the possibility of intentionally excavations, we discovered that the stones were found even under the bushes (see Figure 5).

Figure 5: One of the member (Flavie Isatelle) digging (top), and red andesine from the ground in the same area (bottom).

Furthermore, as we climbed up into the valley until we reached the Yu Lin Gu alluvial fan, the stones were found only on or near the surface at Yu Lin Gu (see Figure 6)

Figure 6: Red andesine samples found on the ground in Yu Lin Gu valley (top), and hand-picked samples from the ground in the same area (bottom).

Even though, the host rock or primary source of these andesine samples could not be found during our expedition, our investigation proved that red andesine does occur naturally at Zha Lin. We could not verify the authenticity of the Yun Lin Gu occurrence because the stone can not be found in pits dug below the surface. Unfortunately, because of a religious dispute, permission to access the mining area above Nai Sa village in Bainang county was denied. Bainang County was previously reported to be the first Tibetan mining area of this gemstone (Abduriyim, 2009)

The quality of andesine roughs found in this area varies from translucent to transparent with rounded edge and colours range from deep red, orange-red to pale reddish orange (see Figure 7). Of course, a few pieces show greenish core with red rim or red zoning. The size of crystal rarely exceeds a cm in length and very few over 2 cms. The crystal surface commonly shows etched features which seem to be more pronounced in samples collected from Yu Lin Gu area as compared with those of the Zha Lin samples.

Figure 7: Red andesine samples found in Zha Lin and Yu Lin Gu (deep red on the left, orange-red in the middle and pale reddish orange on the right)

Even though this expedition to Tibet could prove that red andesines do occur naturally, the crucial question in the trade is how the gem testing laboratory can distinguish these natural materials from their treated counterparts. Further analyzes of samples from this source will be conducted by various techniques and compared with the treated samples in order to be able to separate the natural from the treated ones.

Finally it is also worthwhile to emphasize here that the joint expedition and co-operative research team among each member laboratories are the key factors for the success of this project. The research results will therefore be discussed, concluded and published in the international gemological journals soon.

Article by: Thanong Leelawatanasuk

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The 9 July accident on the Qinghai/Tibet highway seemed to have happened instantaneously, a moment in time, when compared to the time for our escape from China, the time for the current rehabilitation program, and time for further medical procedures we may have to experience For JDM, damage from the accident was found to be extensive, fractured vertebrae, ribs, clavical, pelvis, knee, and some accumulation of blood between the skull and brain. The prayers and help of many friends are so appreciated and have allowed us to come this far in our journey towards the recovery of our health.

For such accidents, death is a frequent outcome. Our survival was by the grace of God We are reminded of promises in the scripture such as Psalm 91:11,12 and 41:1,2. In our case surely many angels were provided from the time of the accident and throughout our journey.

We are all uncertain what the future might hold but with this experience a new perspective has been forged. Each moment in time is viewed as being miraculous. Aren't we all living on borrowed time ? (Scripture). Remember treat your body like you will live forever and your soul like you will die tomorrow.

J.D. Miller
Travel to China
2018 P R China International Science & Technology Cooperation Award
5 – 13 January 2019

Saturday, 5 January 2019: Departed from SLC to China on Saturday, 5 January 2019, with Dr. Xuming Wang. Had a 4-hour lay-over in Seattle, then to Beijing, nonstop on DL 129. Lost my glasses during the night; the stewardess found them buried in the reclining seat, but the lens had come out.

Sunday, 6 January: arrived in Beijing, met by Professor Shenghui Guo (KUST) and off to Friendship Palace Hotel.

Monday, 7 January: Professor Guo reviewed instruction for the ceremony on Tuesday at the Great Hall. Breakfast in the morning, then met Hao Du for lunch at Friday's. I got grilled salmon, Xuming had ribs, onion rings, french fries. Evening banquet at Friendship Palace – host was Huang Wei, Vice Minister, Ministry of Science and Technology. Vivian was the translator for VP Liu, who spoke after the ceremony.

Tuesday, 8 January: Ceremony day, and my instructions were repeated several times. They wanted to make sure I knew what to do. Professor Guo made sure that I was where I was supposed to be during the ceremony. Lots of photos. [JDM should add much more about the ceremony!!!] Chinese lunch, and then to the airport for flight to Kunming with people from KUST, Professor Guo, Vice President Shu, and Director Jian. Received at the Kunming airport by Dianwen Liu, Dean of the Graduate School, and his assistants, with a bouquet of white roses. Then to the campus hotel, the Wenhui Hotel. We have stayed here on previous trips.

Wednesday, 9 January: Breakfast at the hotel, then special symposium at KUST celebrating the ISTC award that I had received in Beijing. The symposium included several speeches from KUST administrators (Vice President Shu Hongshun, Dean Dianwen Liu, and Yunnan DST Director Guan), as well as comments by Xuming and me. Also included in the symposium at KUST was the power point presentation used for my nomination and eventual selection for the ISTC award. Finally, they showed a CCTV video of the Great Hall ceremony. After the KUST symposium, we had lunch, followed by a restful afternoon, and concluding with a western dinner by Lake Dian, south of Kunming. The lake is quite an attraction in the winter with thousands of seagulls migrating from Siberia each winter, seeking better accommodations, only to return in March. Xuming left for an evening with his parents in Dali after the symposium. After Xuming returns we will prepare for departure for Taiyuan, and Shanxi University.

Thursday, 10 January: Breakfast with Professor Shenghui Guo at the Wenhui Hotel on the KUST campus. After breakfast, worked in the hotel room, went for a walk through KUST campus with Shenghui Guo, and then met Xuming for lunch at the hotel. The ISTC award decorated the KUST metallurgy building. Later in the afternoon we were driven to the CYMCO Research Institute on a new campus 40 km south of Kunming and were greeted by about 10 former researchers/students who had been with us at Utah. Jiansheng gave a gift of ceramics

commemorating 60 years of the Institute. On return to Kunming we stopped for dinner, and then, finally returned to the Wenhui Hotel.

Friday, 11 January: After breakfast at the Wenhui Hotel we packed our luggage and left campus for the airport and the flight to Taiyuan, Shanxi. Dianwen Liu had brought gifts for Patricia. Shenghui Guo had brought tea. We were driven to the airport by Xia. The flight to Taiyuan on China Eastern arrived at about 3:00 p.m. Greeted with flowers by Enze and others from Shanxi University where, unexpectedly, honored with a special symposium in celebration of the International Science and Technology Cooperation Award. The symposium had speeches by several administrators, including the President of Shanxi University, and my response. Formalities concluded with the President making a special appointment as XXXXX.

After the symposium, selected faculty and students presented poetry and song for our entertainment. The program is attached together with the poetry and songs that had been prepared.

We then left the Shanxi University campus for our hotel, the Lihua Grand Hotel, a 5-star hotel with luxurious accommodations. The suite provided for me was larger than the first floor of our 1886 Atkin Avenue house in Salt Lake City. We went to the 24th floor restaurant for dinner. I had an excellent salmon dinner with rice and salad. Dessert was vanilla ice cream.

Saturday, 12 January: Excellent breakfast in the morning and then did email and writing in my room while Xuming met with Shanxi researchers. During the morning I was visited by Professor Dong and her students from Taiyuan University of Science and Technology. We discussed some coal research, particularly coal washability analysis by XCT. Professor Dong left a gift of wild tea (natural tea, boil for ½ hour). Xuming and Fangqin Cheng returned with others, including Jimmie, her husband, and we went for lunch. It was at this time I met Fangqin Cheng's granddaughter, You Fi, and her parents. See text message.

Fangqin Cheng – tea set

Sunday, 13 January: After dinner with Hao Du, his wife, Liping, son, Jason, and two former student visitors (Biao XXX and XXX) on Saturday, we stayed a night in a hotel near the Beijing Airport. Sunday morning breakfast was followed by taking the shuttle to the airport. Some hours later, after checking in for boarding passes and a rest in the VIP lounge, we left at about 12:30 p.m. on DL 128 for Seattle.